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Online creators' strategic cooperation in two-sided synchronous UGC market: Empirical evidence from a livestreaming platform

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ABSTRACT

This study explores widespread cooperation among live content creators on synchronous user-generated content (UGC) platforms. It introduces a typology of online cooperation – hyping, peering, and spreading – based on creators' relative popularity statuses and proposes a framework exploring resource-based cooperative strategies and the economic impacts of the three cooperation types. The study reveals varying effects of homophily-based and reciprocity-based strategies on the three cooperation types. The findings also highlight the network effects arising from cooperation and identify conflicting impacts of hyping on creator and platform growth. These findings offer essential insights for managing platform growth in the synchronous UGC market.

1. Introduction

The prevalence of user-generated content (UGC) has reshaped the audience's information-seeking behaviors, shifting focus from official sources to authentic voices. This shift has elevated the market value of UGC, bringing great profitability to marketers and UGC platforms. As a result, UGC platforms have introduced various incentive mechanisms to reward content creators for their contributions and their ability to attract and retain audience. In particular, synchronous UGC platforms, such as Twitch.tv and Douyu, facilitate live and on-demand content contributions, including live video streaming, while enabling synchronous content consumption among audience [1]. These platforms offer creators diverse monetization channels (e.g., live auctions and sales, gifting, virtual currencies such as Bits on Twitch.tv), making live content creators a lucrative profession. Top live content creators today earn six-figure annual incomes [2]. However, content monetization has not only blurred the boundary between leisure and work but also escalated competition among live content creators. To be profitable, these creators must possess a well-structured online profile, a prominent network position, and influential online activities on synchronous UGC platforms [3]. Consequently, live content creators must vie for influence, such as increasing follower or subscriber counts, to boost their visibility [4].

Interestingly, despite intensified competition on synchronous UGC platforms, live content creators often cooperate with their peers, even at the risk of losing their audience to their cooperative partners. For example, streamers on Twitch.tv (henceforth, Twitch), a leading U.S. livestreaming platform, often engage in unique cooperative behaviors specific to synchronous UGC platforms. These behaviors include *hosting* or *raiding* other streamers' live content, effectively redirecting their viewers to the channels of their chosen collaborators. Such cooperation may occur between a pair of live content creators with similar popularities, termed "**peering**," and between two live content creators with distinct popularities, termed "**hyping**"[c] or "**spreading**" (Table 1).

However, the underlying motivations driving cooperation with their potential competitors, which seemingly defy self-interest, remain unexplored. Therefore, this study seeks to answer:

RQ1. What motivates live content creators to participate in the different types of cooperation (i.e., hyping, peering, and spreading) on a synchronous UGC platform?

Furthermore, cooperation with competitors may either cannibalize or expand the market share of live content creators who initiate cooperation. On the one hand, initiating cooperation, such as hosting or raiding others' live content, exposes the initiators' audience to the

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[c] On livestreaming platforms, it is a common practice for less-popular streamers to initiate cooperative events with more popular others to hype up (or self-promote) their own channels. Therefore, we name such cooperative events "hyping events."

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Table 1
Typology of cooperative events on synchronous UGC platforms.

		Initiators (i.e., focal live content creators in this study)	
		Less popular	More popular
Recipients	Less popular	Peering	Spreading
	More popular	Hyping	Peering

content and communities of the recipients. This increases the risks of audience diversion, particularly among dissatisfied, less loyal, or resource-constrained audience [5]. On the other hand, cooperative efforts with other creators may yield positive outcomes for the initiators. For instance, initiating cooperation with others may foster a cooperative reputation and facilitate reciprocal benefits, enhancing the initiators' market performance. Cooperation with others may also offer opportunities for social learning, thereby strengthening the initiators' production capabilities and competitive advantages in the market [6]. Yet, the nuanced effects of the three cooperation types – hyping, peering, and spreading – on initiators' market performance are unclear. Hence, it is imperative to investigate whether the three types of cooperation cannibalize or expand the initiators' market performance and to what extent. Hence, we also investigate:

RQ2. How do the three types of cooperation influence the market performance of live content creators who initiate cooperation[d] on a synchronous UGC platform?

Drawing on the theoretical lens of a resource-based view of strategic cooperation [7] and network effects (NEs) in two-sided markets [8–11], we develop an overarching framework for advancing our knowledge of the strategic considerations underlying live content creators' cooperative behaviors and their economic impacts. With a unique Twitch dataset containing cooperative activities (i.e., hosting and raiding) of 1622 active English-speaking streamers' channels across six months, our empirical results shed light on the cooperative dynamics and their economic impacts on both live content creators' market performance and platform growth in the synchronous UGC market.

Our findings identify four resource-based cooperative strategies (i.e., vocational homophily, structural homophily, direct reciprocity, and indirect reciprocity) as the key drivers of strategic cooperation among live content creators. The findings suggest that vocational homophily consistently emerges as the most salient driver for driving all three types of cooperation. This aligns with prior literature highlighting individuals' tendency to form social ties primarily based on the similarity in their content specialties [12,13]. Moreover, direct reciprocity appears to be the least important for driving upstream cooperation (or hyping), while structural homophily holds the least importance for motivating peer cooperation (or peering). Furthermore, our findings indicate that both hyping and spreading events exert positive NEs on initiators' subscriber gains and follower gains, suggesting expansion effects of hyping and spreading at the creator level. However, our additional analyses revealed that the prevalent hyping events could be detrimental to live content contributions on synchronous UGC platforms.

Our study provides several notable contributions. First, the study contributes to the UGC literature by filling a critical research gap that addresses the incentives to cooperate in the competitive synchronous UGC market. Our findings show the differential impacts of live content creators' resource-based considerations over homophily and reciprocity on the three types of cooperation. Second, this research contributes to the body of knowledge on NEs by identifying the differential NEs that stem from cooperation among live content creators of varying popularity statuses. Beyond theory, we provide implications for synchronous UGC platform stakeholders to strategically promote cooperation and avoid the unintended consequences of cooperation.

2. Theoretical foundations

2.1. Literature on synchronous UGC and livestreaming platforms

A burgeoning body of research on synchronous UGC platforms investigates how social and economic factors impact live content creators' continuous content contributions and their channel growth. Prior literature on livestreaming platforms, for example, has shown that streamers are typically motivated to start their professional streaming careers by the expected financial rewards and the opportunities for intimate interactions with their audience [14]. Moreover, both intrinsic motivations (e.g., challenge seeking, task enjoyment, desire for self-worth) and extrinsic motivations (e.g., expected financial rewards, social capital, peer feedback) are found to influence streamers' perceived performance expectancy, which in turn enhances their perceptions of the platform's attractiveness and encourages them to continue their content contributions [15].

Existing studies on synchronous UGC platforms have also explored the audience's participatory motivations. For instance, livestreaming audience is shown to be motivated by utilitarian value, hedonic value, and social value [16]. While utilitarian value (e.g., usefulness) [16,17] and hedonic value (e.g., novelty) [6] are informational values materialized through livestreaming content, social value emerges from the benefits of the social interactions within a livestreaming channel. Through large-scale surveys, both Hilvert-Bruce et al. [16] and Sjöblom and Hamari [6] have found that audience is motivated to engage in behaviors, such as following, subscribing, and gifting, by social values, which include the attainment of new social ties and the enhancement of community belonging [1]. In addition, researchers have found a positive impact of entertainment- and information-seeking motivations on the audience's time spent [15–17], which, in turn, facilitates their financial engagement (i.e., money spent) on livestreaming platforms [17]. Furthermore, reciprocated interactions in a livestreaming channel have been found to have the most substantial impact on audience's willingness to subscribe to streamers' channels, followed by the impacts on other active (e.g., following) and passive (e.g., viewing) engagement behaviors [18].

The audience engagement, however, may exert two countervailing effects on audience retention in a live content creator's channel. On the one hand, with a larger audience base participating in a livestreaming channel, the existing audience may engage more due to the perceived normative value. On the other hand, the increase in audience may also dilute the average interactivity between a live content creator and the audience, impeding the fulfillment of the existing audience's social goals [19]. In line with this view, researchers have found that audiences who prefer less popular channels are more socially motivated than those who prefer more popular ones [16].

Nevertheless, empirical work on live content creators' cooperation is limited, albeit its ubiquity. An exception is Zhao et al. [20], which explores the impact of streamers' cooperation on the recipients' audience participation and identifies a bystander effect among commenters in the recipients' channels. Yet, little is known about why live content creators initiate cooperation with their potential competitors and how their initiated cooperation contributes to their market performance and platform growth. When a creator initiates cooperation with others, the audience of the initiators may derive additional informational value since the content-related cooperation may help audience discover new content with lower search costs [21]. Besides, frequent cooperation may

[d] Our study focuses on the initiators of cooperation for two reasons. First, our study is interested in the motivations and economic impacts of streamers' strategic cooperative decisions. Thus, focusing on the initiators who strategically select their cooperative partners is better aligned with our research goals. Second, prior livestreaming literature has explored the impact of cooperation on the recipients [55]. We build our study upon but differentiate our study from the prior literature.

enhance the initiator's cooperative reputation and lower other creators' cooperative uncertainty [22]. Live content creators who often cooperate with others may also be more informed about the content trend and audience's needs [23], thereby producing market-driven content that audience finds most useful or novel. Given that cooperation initiated by live content creators may not only influence the creators' contribution motivations intrinsically and extrinsically but also affect the channel utilities to the audience, this study proposes a research framework that helps develop a nuanced understanding of the cooperative dynamics on synchronous UGC platforms.

2.2. Cooperation strategies

Cooperation, in general, involves individuals incurring a cost to interact with and help others achieve their goals [24]. In online communities, for example, it is not uncommon for users to spend time sharing valuable information or endorsing other peers to show their support. Online users may also monetarily invest in others' business proposals or grant other users access to services or products. Although cooperative behavior could be purely altruistic and strategy-free [10], this idealistic view seems less explicable in the real world, where intense competition prevails. For instance, in the online content market, with numerous content suppliers vying for the attention of a limited audience base, forming cooperative relationships often transcends altruistic motives and becomes strategic [10].

Literature on cooperation has so far investigated self-interested cooperative strategies based on homophily and reciprocity to explain this altruism-like behavior [25]. Homophily, rooted in social identity theory [26], illustrates individuals' tendency to cooperate with someone alike primarily based on their demographical, structural, and vocational identities [27]. While demographical identities (e.g., age, gender) are categorized as ascribed characteristics, structural and vocational identities are categorized as achieved characteristics [28]. Since engagement on synchronous UGC platforms does not require live content creators to reveal their actual personal profiles, these creators are more likely to be grouped based on their online visibility, team memberships, and content specialization rather than their demographic attributes. Thus, this study focuses on achieved characteristics, namely structural and vocational homophily.

Structural homophily describes individuals similarly embedded in a social network with shared structural constraints [28–30]. For instance, live content creators belonging to the same team share a common social circle and similar structural properties. Cooperating with structurally homophilous others can be a strategic means to enhance and safeguard one's social relationships. By cooperating with others from the same social circle, individuals reinforce their social identity within the group [26], leading to preferential treatment, support, and recognition from fellow members. This, in turn, solidifies one's relational capital within the group [31]. *Vocational homophily* reflects the similarity in specialty (e.g., content specialty) between two individuals [28]. Cooperating with vocationally homophilous others can facilitate access to consumers aligned with the initiators' areas of expertise, attracting a larger consumer base. It can also promote relevant knowledge sharing and learning, thereby preserving and enhancing initiators' reputation within their knowledge domains [32].

Both structural and vocational homophily may coexist on multilayered synchronous UGC platforms, where social interactions span across various layers of social circles formed by different identities [33]. For instance, a live content creator's identity is shaped by both what team(s) the creator belongs to and what content the creator frequently streams.

Reciprocity is another resourced-based strategy that plays a major role in forming a cooperative relationship [10,25]. Reciprocity can be direct or indirect. *Direct reciprocity* occurs in cooperative events where individuals have cooperative experiences with the selected partners and initiate cooperation in return for their previous favors (i.e., giving back). The norm of direct reciprocation plays a central role in facilitating dyadic cooperation as it helps stabilize existing partnerships, promote trust, and strengthen social ties between the two cooperative partners [34]. *Indirect reciprocity*, on the other hand, manifests in first-time cooperative relationships, where individuals do not have cooperative experiences with their selected partners. Instead of directly returning favors to those who cooperated with them, individuals who are driven by indirect reciprocity give back to the community by initiating cooperation with new partners [22]. The concept of indirect reciprocity extends cooperative behaviors beyond direct reciprocal relationships to a broader network. It enables individuals to tap into new consumers and partners through novel cooperative relationships.

Both direct and indirect reciprocity can be the salient factors driving cooperation among live content creators on synchronous UGC platforms. This is because the self-interested cooperation among live content creators is a social exchange process in which creators cooperate in exchange for either instrumental (e.g., audience, relationships) or symbolic benefits (e.g., reputation) [35]. Direct reciprocation to other creators' previous favors could help live content creators maintain and reinforce their existing cooperative relationships. Indirect reciprocation benefits live content creators by facilitating creators' access to new partners and expanding their audience base at the community level.

2.3. Network effects and value cocreation on two-sided synchronous UGC platforms

Two-sided markets [36,37], such as eBay, crowdsourcing platforms, and UGC platforms, are characterized by both same-side NEs [38] and cross-side NEs [39]. Positive same-side (cross-side) NEs occur when users derive values from each unit increase in the installed base or usage on the same market side (the other market side), whereas negative same-side (cross-side) NEs occur when each unit increase in the installed base or usage hurts the platform value to users on the same side (the other side).

Like other UGC platforms, synchronous UGC platforms are essentially two-sided markets with live content creators and audience on the two sides. The positive or negative NEs of a synchronous UGC platform result primarily from the exchange among platform users [5,40]. Specifically, each additional user adds or diminishes platform value to other users through their interactions with others on the platform. From the value cocreation perspective, the added or diminished platform value can be *value-in-exchange, value-in-use*, or *value-in-experience* [41]. *Value-in-exchange* is an output-based value often referring to financial (e.g., profitability) and social outcomes (e.g., social ties and reputation). *Value-in-use* is a process-based value corresponding to the perceived personal integrative value, such as knowledge and competency, resulting from using the product. *Value-in-experience* involves individual sensemaking, which is subjective based on the perceived functional (e.g., product quality and usefulness), hedonic (e.g., novelty and enjoyment), and spiritual values (e.g., emotional support and sense of community) of the product [41].

The exchange among platform users may happen on the same market side (i.e., either the creator side or audience side) or across the two sides. Cooperation between live content creators is the same-side exchange that is likely to affect the value that the platform offers to these creators. This is because such cooperative events may influence 1) creators' reputation and profitability through sharing audience bases (i.e., *value-in-exchange*), 2) creators' knowledge and productivity through learning from their cooperative partners (i.e., *value-in-use*), and 3) perceived emotional support to the creators (i.e., *value-in-experience*). The interactions among audience (e.g., chatting with or following each other) are also same-side exchanges that enhance the platform value to the audience as such interactions may expand the participants' social circles (i.e., *value-in-exchange*) and facilitate a strong sense of community (i.e., *value-in-experience*). More intuitively, the cross-side interactions between live content creators and their audience may affect the platform value exerted on each other. On the one hand, live content creators

deliver *value-in-experience* to their audience through quality and novel live content and by rewarding their support and participation. On the other hand, audience engagement affects the *value-in-exchange* of live content creators as the increases in viewership and followership promote creators' ad revenues and the likelihood of being subscribed [18].

Cooperation among live content creators may influence the utilities derived from both the same-side and cross-side exchanges mentioned previously, thus altering the same-side and cross-side NEs. This study, therefore, hypothesizes the differential NEs stemming from creators' cooperation.

3. Research framework and hypotheses development

We present our research framework (Fig. 1) and develop our hypotheses as follows.

3.1. Resource-based strategic cooperation on synchronous UGC platforms

The resource-based view of strategic cooperation suggests that strategic cooperation serves as a market strategy designed to optimize market returns by pooling and leveraging valuable resources from cooperative partners [7]. This perspective highlights two intertwined motives for market suppliers (e.g., content creators) to pursue strategic cooperation: (1) accessing additional resources that are otherwise unattainable in isolation [42] and (2) retaining one's own resources [43, 44]. The former prioritizes growth and expansion, whereas the latter emphasizes preserving the existing status quo [7].

In the synchronous UGC market, the success of a live content creator hinges on three key factors: audience influence, network position, and reputation [3]. Accordingly, resources on a synchronous UGC platform can be categorized into three categories: (1) *audience resource,* denoting the level of engagement and interaction that live content creators foster with their audience; (2) *relational resource*, referring to the network of social relationships that live content creators establish with their peers; and (3) *reputational resource,* reflecting the credibility and acknowledged expertise of live content creators [35]. Live content creators often engage in strategic cooperation to either (1) access additional resources, prioritizing resource expansion, or (2) retain their own resources, emphasizing resource retention. Homophily-based and reciprocity-based cooperative strategies can enable live content creators to selectively access or retain these distinct resources, aligning with their strategic goals.

On a synchronous UGC platform, live content creators may stream different content over time but devote most of their streaming efforts to one or two contents they feel most competent at or comfortable with. Therefore, the frequently streamed content(s) reflects a creator's content specialty. In such a context, live content creators are more likely to initiate cooperation with other creators specializing in the same content (i.e., vocational homophily), as this form of cooperation can be instrumental in enabling their access to audience resources that are aligned with their area of content expertise. Moreover, cooperation with other creators who share the same content specialty fosters the exchange of relevant knowledge and learning. Such cooperation augments the initiators' acumen within their respective knowledge domains and enhances their reputational resources [32]. Owing to these benefits, live content creators are likely to establish strategic cooperation with others who possess expertise in the same content areas. For example, on Twitch, streamers specializing in League of Legends often collaborate with other streamers who focus on the same game by hosting or raiding them (practices where one streamer directs their audience to another's live stream at the end of their sessions).

Furthermore, such strategic tendency may be stronger in hyping events, compared to peering and spreading events. This is because hyping events offer a unique opportunity for live content creators to leverage the reputational and audience resources of their higher-status partners. In these cooperative events, the strategic alignment in content can significantly boost the lower-status initiator's visibility and audience reach, which might not be as pronounced in peering or spreading events where the status differential and audience size may not be as substantial. Thus, we hypothesize that:

Hypothesis 1. Vocational homophily is positively associated with the number of cooperative events initiated by live content creators (H1a). Such a positive effect is more salient in hyping events, compared to peering and spreading events (H1b).

Unlike the preferences to pursue vocational homophily, live content creators' preferences for cooperating with others who are structurally proximate (i.e., structural homophily), such as those who belong to the same team, are likely to be driven by normative pressure [45]. Normatively, live content creators who function as teammates are expected to adhere to team norms and beliefs, driven by shared team values that foster a cohesive group identity [46]. The homophilous team structure on synchronous UGC platforms typically encourages creators of the same team to collaborate with each other to facilitate individual growth and improve the aggregated team welfare. The violation of team norms may result in team punishment (e.g., being kicked out or isolated), thus potentially harming the existing social relationships and reputation of the initiators. Consequently, to maintain existing relational and reputational resources, live content creators are more likely to initiate cooperation with team members. For instance, on Twitch, streamers affiliated with professional esports organizations or content creator collectives – such as TSM or OfflineTV –frequently collaborate on team events, tournaments, or charity streams. Moreover, we propose that structural homophily exerts a less pronounced influence on driving peering events compared to its role in promoting hyping and spreading events. This can be attributed to the potential competition arising from peering with similar-status others within the same team, which may jeopardize their shared audience resource [33]. Hence, we hypothesize that:

Hypothesis 2. Structural homophily is positively associated with the number of cooperative events initiated by live content creators (H2a). Such a positive effect is less salient in peering events, compared to hyping and spreading events (H2b).

Practically, like many other UGC platforms, synchronous UGC platforms provide convenient features for live content creators to trace their previous cooperative partners, enabling live content creators to return others' favors directly. Strategically, live content creators may hope to strengthen their social ties with those who are willing to cooperate with them to stabilize their cooperative social circles [34]. By conforming to the reciprocal norm in dyadic cooperative relationships, live content creators may also expect to maintain or enhance their cooperative reputation [35], thereby attracting more cooperative partners. To illustrate, on Twitch, a streamer who is frequently raided or hosted by another creator may return the favor by raiding or hosting the same creator in subsequent streams. As a result, live content creators may directly reciprocate others' previous favors to strategically maintain their existing relational resources as well as access new partners through enhanced reputation. We also posit that the effect of direct reciprocity can be more pronounced in peering events as opposed to hyping and spreading events. This is because cooperation with others of similar status is often driven by individuals' expectations of more balanced and equitable exchanges of resources [47], in addition to maintaining and accessing resources. Thus, to foster equitable resource exchange, live content creators in peering events can be more inclined to reciprocate directly with those they cooperated with, as direct reciprocal relationships are more stable and trustworthy [34]. With these reasonings, we hypothesize that:

Hypothesis 3. Direct reciprocity[e] is positively associated with the

[e] Our study focuses on ex-post reciprocity (i.e., giving back), which is distinct from ex-ante reciprocity (i.e., anticipated reciprocity) [70].

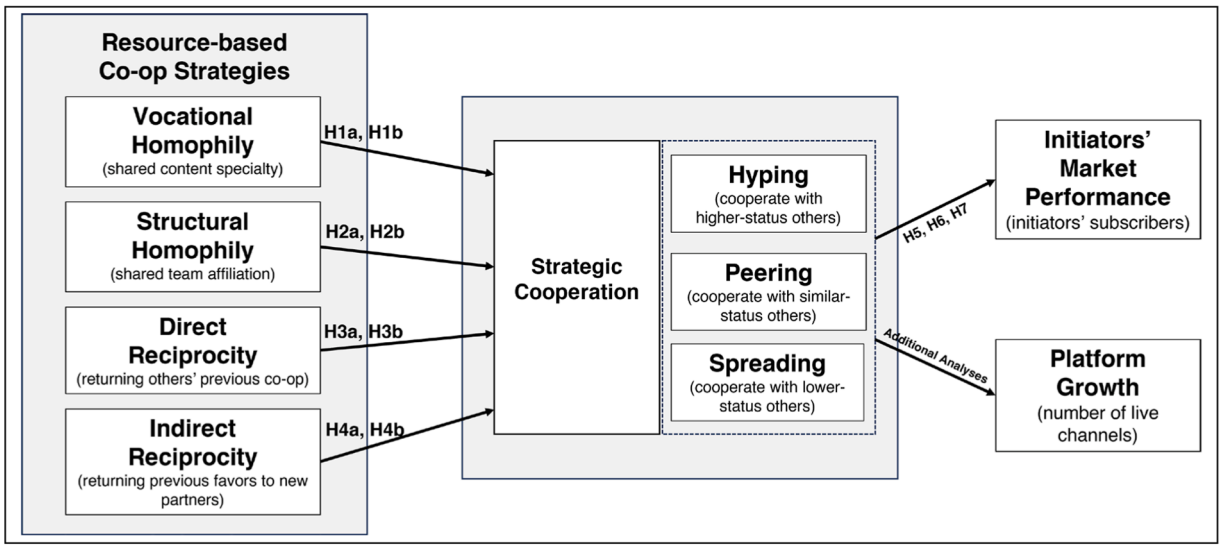


Fig. 1. Conceptual framework.

number of cooperative events initiated by live content creators (H3a). Such a positive effect is more salient in peering events, compared to hyping and spreading events (H3b).

Beyond the direct dyadic reciprocal exchange between two live content creators, reciprocity may occur indirectly at a broader scope on a synchronous UGC platform. In other words, live content creators may not only return the favor to those who have helped them but also give back to the community by cooperating with new partners [22]. Indirect reciprocation by initiating cooperation with other first-time partners is likely to help initiators extend their cooperative circles by forming new cooperative relationships and spreading their cooperative reputation in the network. This would, in turn, enrich live content creators' relational resources and help them gain access to new audience pools. For example, on Twitch, a streamer who has benefited from community raids, subscriptions, or donations may choose to collaborate with or raid an emerging streamer they have not previously worked with. Hence, we propose indirect reciprocity as a resource-based strategic driver for live content creators' cooperation on a synchronous UGC platform. Furthermore, we propose that indirect reciprocity plays a more significant role in facilitating hyping events, as compared to its influence on peering and spreading. Hyping involves interactions where lower-status individuals actively seek the attention and favor of those with higher status, aiming to expand their own networks and visibility. In these events, the lower-status individuals can gain significant reputational benefits by being associated with higher-status peers. This association can lead to broader recognition within the community, which is crucial for their professional growth and visibility. Specifically, we hypothesize that:

Hypothesis 4. Indirect reciprocity is positively associated with the number of cooperative events initiated by live content creators (H4a). Such a positive effect is more salient in hyping events, compared to peering and spreading events (H4b).

3.2. Network effects (co-created values) of cooperation at the individual level

Drawn from the theories of NEs and value cocreation, we define the positive (negative) network effects of cooperation as the NEs arising from the potential values added (diminished) through the increase in cooperative events among live content creators on a synchronous UGC platform. We theorize the NEs of cooperation at the individual level based on the three co-created values – *value-in-exchange, value-in-use*, and *value-in-experience*. As reviewed in Section 2.3, *value-in-exchange* is an output-based value that often refers to financial and social outputs during exchanges. *Value-in-use* is a process-based value referring to the personal integrative value, such as knowledge and competency, resulting from using the product. *Value-in-experience* involves individual experience and sensemaking based on perceived functional, hedonic, and spiritual values of the product [41].

We propose positive NEs of live content creators' hyping events on the initiators' market performance for several reasons. First, hyping events offer invaluable learning opportunities for less-popular creators. By engaging with more-popular counterparts, more-popular creators are exposed to advanced content production techniques and gain insights into varying audience preferences and market trends. This exposure is instrumental in enhancing their content quality and market knowledge, ultimately elevating their *value-in-use*. Evidence in online contests supports this notion, where individuals engaging with "superstars" in the market often experience accelerated learning curves, acquiring crucial skills that contribute to their success [48]. Moreover, such cooperation can lead to the discovery of new audience segments for less-popular creators, thereby enhancing their *value-in-exchange*. Furthermore, audience today are increasingly seeking variety and quality in content [49,50]. By aligning with more-popular creators, less-popular individuals can leverage this trend, transforming their channels into hubs of diverse and high-quality content. This, in turn, levels up the *value-in-experience* of audience. For these reasons, we hypothesize that:

Hypothesis 5. On a synchronous UGC platform, a live content creator's frequency of initiating hyping events is associated with positive NEs on the initiator's market performance.

We suggest that positive NEs will arise from live content creators' cooperative events with other creators of similar popularity statuses (i.e., *peering* events) on a synchronous UGC platform. In dyadic cooperative relationships, individuals are more likely to share similar experiences and value perspectives with similarly popular partners, compared to others with heterogenous popularity (i.e., more or less popular). Therefore, peering with others may not only validate the attitudes and values of those who are involved but also lead to mutually positive evaluations between the two cooperative partners [51]. This further promotes affective closeness and trust (*value-in-experience*) as well as strengthens the social ties (*value-in-exchange*) between two status-equivalent individuals [28]. The enhanced affective and social capital may then motivate the creators involved in a peering event to contribute live video content more frequently and with higher quality,

which helps them acquire more audience [15].

Furthermore, audience of two live content creators with similar popularity statuses are likely to share common preferences in selecting live channels to watch. For example, audience of less-popular channels may prefer more interactions with the creators of the channels, while audience of more-popular channels may enjoy socializing with other audience in the community or simply being bystanders [16,20]. The similar popularity of creators also signals to the audience the similar content quality [52] and content experience. Hence, peering with other creators of similar popularity could enhance the *value-in-experience* of the initiators' audience by directing them to other channels that they would find enjoyable while lowering their search costs in locating new content. Additionally, cooperating with other creators of similar popularity may also help the initiator learn more about relevant content production skills (*value-in-use*) that could be effectively applied to attract audience with common preferences. With these logics, we hypothesize that:

Hypothesis 6. On a synchronous UGC platform, a live content creator's frequency of initiating peering events is associated with positive NEs on the initiator's market performance.

We believe that live content creators' cooperative events initiated with less-popular creators (i.e., *spreading* events) lead to positive NEs of cooperation on a synchronous UGC platform for the following reasons. First, receiving favors from more-popular others promotes the recipients' viewership, leading to greater content profitability, at least in the short run. In response to the initiators' favor, the recipients may express their appreciation during the streaming or reciprocate the cooperation directly. Frequent reciprocations from other less-popular creators allow more-popular initiators to acquire new audience (*value-in-exchange*), further extending their popularity on the platform.

Second, cooperating with less-popular creators can also enhance the reputation (and thus *value-in-exchange*) of the more-popular initiators. This is realized through indirect returns from creators or audiences who did not directly receive favors from the initiators but appreciate the initiators' generosity and kindness [22]. Such generous and cooperative acts of the more-popular initiators may also lead less-popular creators to perceive a higher probability of being reciprocated if they choose these initiators as cooperative partners. The increases in the indirect returns from other creators and audiences power the more-popular initiators to move up the popularity hierarchy on the synchronous UGC platform.

In addition, creators contribute more live video content when they are recognized by others and perceive self-worth [15]. High uncertainty in content profitability induced by intense market competition often hinders newcomers from joining to contribute content [53]. Spreading events that benefit less-popular creators would not only encourage them to contribute more live video content but also lower the uncertainty for potential creators to join the platform, leading to increased content contributions and content diversity. Consequently, diverse content selection helps attract more audience to the platform, strengthening the *value-in-exchange* of creators on the platform. Compared to the less-popular creators, more-popular creators who possess more resources (e.g., production skills, social capital) are more capable of capturing the benefits from the increase in audience (i.e., Matthew Effect) [52,54]. Therefore, spreading events may contribute to the popularity of the more-popular initiators. Accordingly, we hypothesize that:

Hypothesis 7. On a synchronous UGC platform, a live content creator's frequency of initiating spreading events is associated with positive NEs on the initiator's market performance.

To test our hypotheses, we collected extensive data from Twitch platform and employed mixed methods, integrating k-means clustering to classify creator popularity statuses and econometric models to evaluate the proposed effects. We demonstrate our research methodology as follows.

4. Research methodology

4.1. Research context

Our empirical setting is Twitch, a leading U.S. livestreaming platform founded in 2011 and currently having >2 million average concurrent viewers and approximately 9 million active streamers. Like many other traditional UGC platforms (e.g., Twitter), Twitch offers various social interactive features, such as sharing, instant messaging, and following, enabling communication and collaboration between streamers and audience. Compared to other UGC platforms, competition among Twitch streamers is more intense due to its synchronous nature of content delivery and audience's participation. To retain its popular streamers, Twitch provides them with greater rewards, such as 70 % of monthly subscription revenue (compared to 50 % for less popular streamers). This further contributes to the growing inequality and disparity among streamers [9].

To promote equal growth opportunities for streamers, Twitch has introduced different cooperative mechanisms to facilitate cooperation among streamers. For example, by hosting live content before or after their own streaming, Twitch's streamers can share their viewers with other streamers. They can also co-stream with others or "raid" others by bringing their viewers to the target channels during their streaming. The hosting mode, launched in July 2014 and disabled in November 2022, has the most extended history on Twitch. The raiding mode, implemented in 2017, is currently the most used cooperative feature on Twitch.

The initiators of hosting and raiding events can choose target streamers' channels from the same team they associate with, self-select any channel they would like to host or raid, or randomly match with channels with similar or different average viewership on the platform. Once the hosting or raiding event is initiated, the audience who visits and engages with the initiators' channels will be redirected to the target streamer's channel, contributing to the target streamer's concurrent viewership. The initiator's audience may choose to leave the target streamer's channel at any time or may follow or subscribe to the channel (see Fig. 2a).

Moreover, the target streamer's channel and its audience will be notified of the hosting or raiding event through a live chat box. The target streamer may express appreciation during the streaming, and the audience may be navigated to the initiator's channel. In this way, the initiator's channel may earn extra exposure through the initiated hosting or raiding event. The initiator's account will also be recorded in their target streamers' hosting or raiding histories to be reviewed in the future (see Fig. 2b). However, hosting mode only allows streamers to initiate when they are offline, while raiding mode only allows streamers to initiate when they are actively streaming with active viewers in their channels. The coexistence of competition and cooperation and the pervasive cooperation among streamers make Twitch an exciting and appropriate setting for our study.

4.2. Data collection and sampling method

We first collected account information (e.g., account IDs, twitch IDs, average concurrent viewers, total stream minutes, affiliated team IDs) of 700,013 Twitch English-speaking streamers across 12 months prior to our study period (December 30, 2021, to June 30, 2022) through an aggregated Twitch information site (sullygnome.com[f]). In this study, we focus on dedicated channels whose roles as streamers significantly

[f] Sullygnome.com is an aggregated Twitch information site that uses Twitch API to track statuses of Twitch channels who are Twitch partners, or affiliates with 3 or more viewers, or non-partners/affiliates with >10 viewers. This sampling strategy is in-line with the sampling criteria and does not eliminate the active focal streamers we wish to identify.

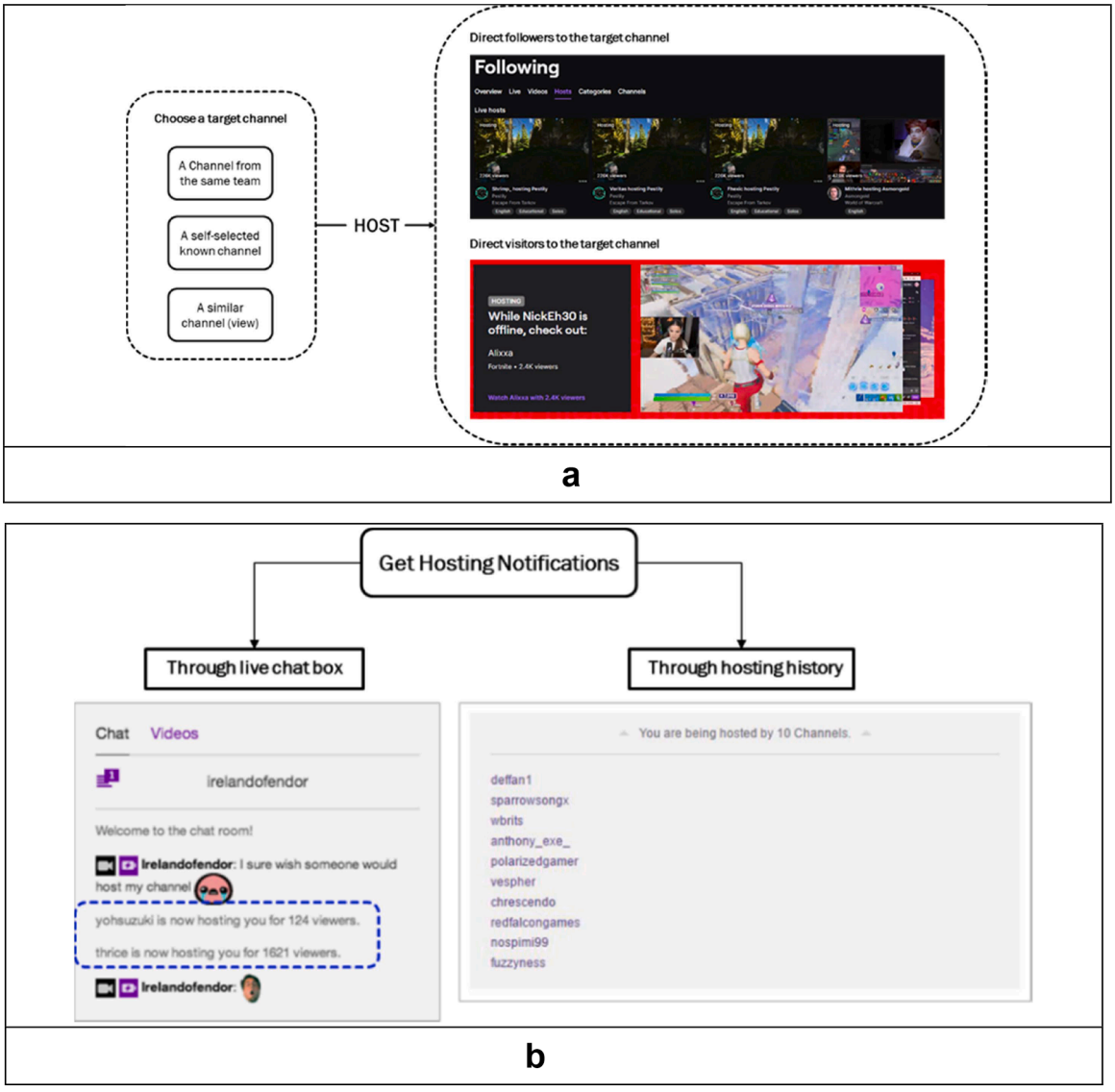


Fig. 2. a. Host initiating a hosting activity on Twitch. b. Hostee getting hosting notifications on Twitch.

outweigh their roles as an audience. Therefore, we filtered out streamers who streamed less than, on average, 20 min per day and had 30 concurrent viewers per stream in order to exclude inactive channels in our samples.[8] This threshold leaves us with a total of 31,896 valid English-speaking streamers.

Given the large number of streamers, we adopted the *k-means* clustering approach. The clustering approach is necessary for identifying distinct groups of streamers based on their popularity status, enabling a more nuanced analysis of cooperation strategies between and across the different segments. Specifically, we employed average viewers and the number of followers as the two key dimensions for clustering. The choice of these dimensions is grounded in the common criteria used by audiences on livestreaming platforms to assess a streamer's popularity [55]. Average viewership provides insight into the regular engagement level of a streamer's content, while the number of followers indicates the breadth of their audience base. This dual-dimensional approach allows for a more nuanced and accurate categorization of streamers, reflecting both the depth and reach of their influence on the platform. This approach allows us to minimize the popularity distance within clusters and maximize the popularity distance between clusters through an iterative procedure [55,56]. As the k-means clustering approach requires a predetermined number of clusters, we used the Elbow method [57]. We iterated the clustering procedure with numbers of centroids (k) from 2 to 10. For each iteration, we calculated the within-cluster sum of squares (WCSS) value (i.e., the variability of the observations within clusters). The optimal number of clusters can then be determined when adding more clusters no longer reduces the WCSS. The resultant WCSS values and their decreasing rates indicate that five is the optimal number of clusters, as the WCSS values start to decrease at a slower rate after five clusters (see Fig. 3). Hence, we clustered streamers into five popularity tiers (see Table 2a). For feasibility, we randomly selected 1000 channels from the Tier 1 streamers, and the distribution is shown in Table 2b.

Based on the selected focal streamers, we acquired detailed cooperative event data that occurred from December 30, 2021, to June 30, 2022, from Esports Charts (https://escharts.com/), a company specializing in livestreaming platform analytics. The comprehensive dataset

[8] The applied thresholds align with Twitch's own benchmarks for active streaming and are supported by empirical statistics. According to the Statista report on Twitch , the average number of concurrent viewers of a streamer on Twitch was approximately 30. Moreover, statistics on Sullygnome also show that 7 million streamers streamed an average of 70 million hours per month on Twitch in 2022, indicating an average 20-minute daily streaming duration per streamer in 2022.

contains hosting and raiding events initiated by the focal streamers, as well as the hosted and raided events received by the focal streamers, with Twitch IDs, event timestamps, and the numbers of followers and subscribers gained during each event of both initiators and recipients of the events. We then collected the streaming content and the team memberships of both focal streamers and their cooperative partners during the study period using Twitch API.

4.3. Key variables and measures

Measures of Initiators' Market Performance. To measure focal streamers' market performance, we specifically focused on the changes in the number of subscribers and followers during cooperative events (hosting or raiding events). Specifically, we collected unique measures of the number of subscribers and followers an initiator gained during his/her initiated cooperative events. These unique measures provide a valuable tool for measuring the direct impact of cooperative events on focal streamers' market performance while controlling for the effects of other events. We calculated the total numbers of subscribers and followers gained from cooperation at the streamer level (*SUBGAIN* and *FOLGAIN*, respectively), and used *SUBGAIN* in our main analysis and *FOLGAIN* for robustness checks.

Measures of Strategic Cooperative Events. A hyping event occurs when a focal streamer initiates a hosting or raiding event with higher-tier streamers. Similarly, peering and spreading events occur when focal streamers initiate hosting or raiding events with similar-tier and lower-tier streamers, respectively. We counted the total hyping, peering, and spreading events initiated by each focal streamer across our study period. As the occurrences of these events are likely to be skewed by streamers' streaming activities, we used the occurrence probabilities of hyping, peering, and spreading per active streaming hour (*HYPE*, *PEER*, *SPREAD*) in our main analyses. Moreover, we also aggregated these measures to measure the occurrence probability of cooperative events per active streaming hour in general (*COOP*). We used the count measures as robustness checks.

Measures of Resource-based Cooperative Strategies. Vocational homophily occurs when a focal streamer initiates a hosting or raiding event with another streamer specializing in the same content. On a synchronous UGC platform, streamers tend to specialize in a limited range of content and rarely stream random and unfamiliar content. Our dataset substantiates this trend, revealing that nearly all streamers devoted over 80 % of their streaming hours to one or two specialized content. The more time a streamer dedicates to specific content, the more likely it becomes recognized as their area of expertise. To determine the content specialty of a streamer, we collected the total streaming minutes as well as the top ten content (based on the length of the streaming minutes) of all the focal streamers and their paired cooperative partners streamed across 12 months prior to our study period. Taking into account the possibility of multi-specialties, we used the top two content[h] as the proxies of a streamer's content specialty. After determining the content specialty of both focal streamers and their paired cooperative partners, we then matched the content specialty of each cooperative pair. We denoted 1 if there is any specialty match between focal streamers and their partners and 0 otherwise. We then counted the total vocationally homophilous events initiated by each focal streamer and calculated the probability for such an event to occur per active streaming hour (*VHOMO*).

A structurally homophilous event occurs when a focal streamer initiates a hosting or raiding event with another streamer who shares the same team membership(s). A team on a synchronous UGC platform like Twitch has its own landing page that gathers a group of streamers to collaborate and promote each other's channels (see Fig. 4). It serves as a collective identity under which individual streamers can unite, often with a shared goal (e.g., esports branding, community building). We then measured structural homophily by counting the total structurally homophilous events initiated by each focal streamer and calculated the probability of their occurrence per active streaming hour (*SHOMO*).

To a focal streamer's direct reciprocity, we used 1 to denote if a focal streamer directly reciprocated a hosting or raiding event initiated by another streamer within 30 days and 0 otherwise. We define a 30-day reciprocal window as the direct reciprocity decays significantly after 30 days, as suggested by Chuan et al. [58]. For robustness checks, we extended the reciprocal window to 60 days. We counted the total direct reciprocity for each focal streamer and calculated the probability for direct reciprocity to occur per active streaming hour (*DREC*).

The concept of indirect reciprocity implies that a focal streamer reciprocates previously received favors indirectly by engaging in cooperative events with new partners within a broader network [22]. Therefore, to quantify a focal streamer's level of indirect reciprocity, we first counted the occurrence of the focal streamer's cooperative events with other streamers who have not previously cooperated with the focal streamer. We then normalized the measure by dividing the count by the total number of streaming hours logged by each focal streamer across our study period (*INDR*).

Control Variables. To model the cooperative strategies behind strategic cooperation, we controlled for other attributes of focal streamers, such as their genders (*GENDER*), their popularity tiers (*TIER*), and the number of days since they created their Twitch accounts (*TENURE*). We also included the average viewership of each focal channel's specialized content(s) to control for a potential content-level influence (*avgSPECPOP*). Moreover, to control for a focal streamer's tendency to cooperate with more popular others [59], we controlled for the cooperative partners' average viewership during the hosting and raiding events (*avgR_HOSTVIEW* and *avgR_RAIDVIEW*). Furthermore, we controlled for the interaction effects between the vocational homophily and structural homophily, as prior studies suggest that multidimensional homophily may alter competitive dynamics and impact individuals' strategic decisions [11,33]. To model the NEs of cooperation, we additionally controlled for partners' average subscriber gains from both hosting and raiding events (*avgR_HSUBGAIN* and *avgR_RSUBGAIN*) and the total streaming length of each initiator measured in minutes (*STRMLEN*). We present the summary statistics and correlation matrix of the key variables in Table 3 and Appendix E Table E1, respectively.

4.4. Model specifications

4.4.1. Cooperative strategy models

To test our cooperative strategy hypotheses (H1-H4), we adopted the Fractional Logistic Response Model (FLRM) [60,61] as our dependent variables, *HYPE*, *PEER*, and *SPREAD*, are fractional variable bounded between 0 and 1. In the case of fractional dependent variables (DVs), FLRM can outperform linear regression models for three reasons. First, linear regression models can be biased when applied to fractional DVs as they allow DVs to fall outside of the range of 0 and 1. Second, FLRM accounts for the floor and ceiling effects that are inherent to bounded fractional variables [60], while linear regression models do not. Moreover, FLRM can be performed without ad hoc data transformations, allowing a straightforward estimation of the conditional expectation of the DVs based on the independent variables [62]. Given these reasons, a general specification of our cooperative strategy models is as follows:

$$E(Y_i|X_i) = Z(\beta X_i),\ Y_i \in [0,\ 1],$$

where Y_i is the DV (*COOP*, *HYPE*, *PEER*, or *SPREAD*) for initiator *i*. X_i is a vector of our independent variables (*VHOMO*, *SHOMO*, *DREC*, and

[h] We used top two content to measure a streamer's content specialties only if the streaming minutes of the top second content surpasses 12 percent of the total streaming minutes of the streamer, which is the mean percentage of total streaming minutes of all streamers' top second content. Otherwise, only the top first content was used to measure a streamer's content specialty.

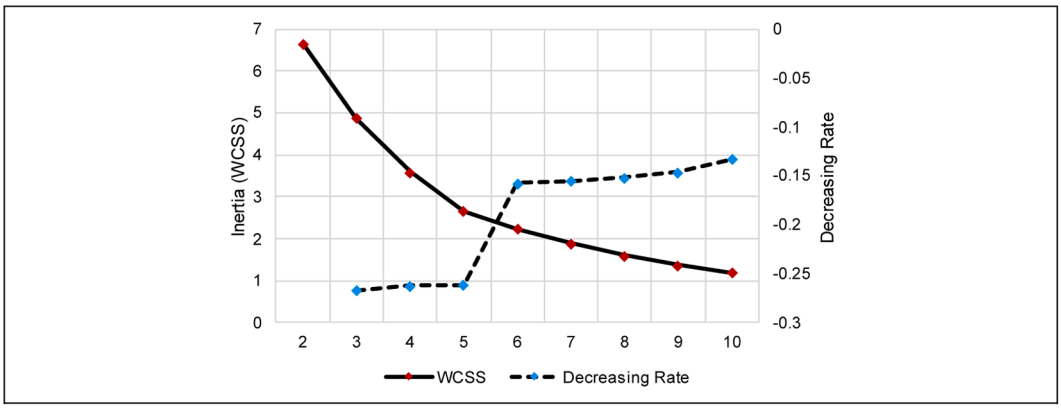


Fig. 3. Elbow method for optimal number of stream channel clusters.

Table 2a
Clustered English-speaking channels.

Tiers	Count	Mean, Avg CCV	Std. Avg CCV	Min. Avg CCV	Median, Avg CCV	Max, Avg CCV
1	31,274	149.673	294.605	30.000	63.000	5830.000
2	529	4117.329	3287.000	30.000	3134.000	20,307.000
3	75	21,543.533	12,293.709	2043.000	19,202.000	61,325.000
4	12	29,727.250	20,781.249	1486.000	29,553.000	74,931.000
5	6	106,054.333	44,739.301	64,677.000	85,370.000	164,429.000

Note: Avg CCV = Average Concurrent Viewers.

Table 2b
Selected English-speaking channel samples.

Tiers	Count	Mean, Avg CCV	Std. Avg CCV	Min. Avg CCV	Median, Avg CCV	Max, Avg CCV
1	1000	144.192	270.433	30.000	41.000	610.000
2	529	4117.329	3287.000	30.000	3134.000	20,307.000
3	75	21,543.533	12,293.709	2043.000	19,202.000	61,325.000
4	12	29,727.250	20,781.249	1486.000	29,553.000	74,931.000
5	6	106,054.333	44,739.301	64,677.000	85,370.000	164,429.000

Note: Avg CCV = Average Concurrent Viewers.

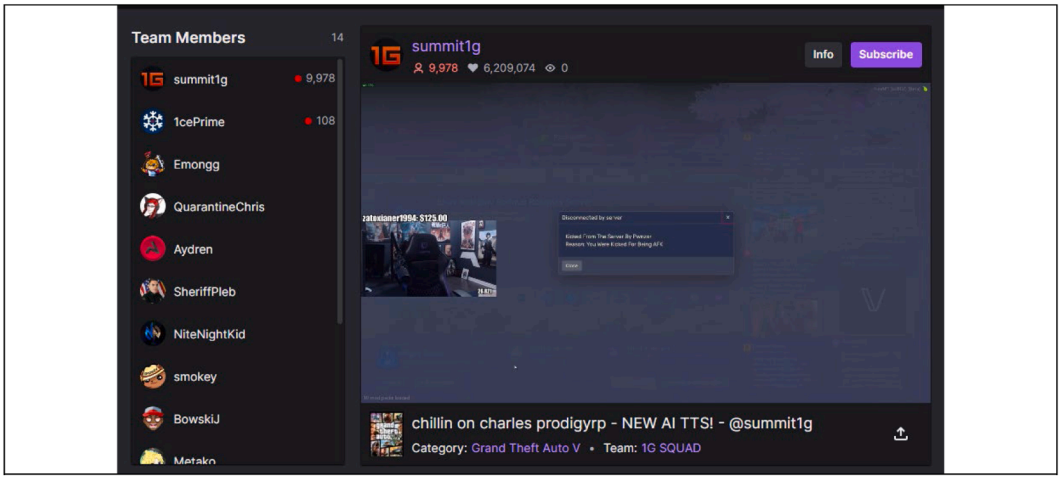


Fig. 4. Example of Twitch team.

INDR), the interaction term *VHOMO* × *SHOMO*, and other control variables for initiator *i*. We excluded focal streamers who belong to Tier 5 when estimating *HYPE*, and excluded those who are in Tier 1 when estimating *SPREAD*. β is a vector of odd ratios to be estimated. $Z(\cdot)$ is a cumulative distribution function that follows a logistic distribution function. The FLRM model is estimated through quasi-maximum likelihood estimation (QMLE), where the likelihood for an observation is explicitly defined as the Bernoulli likelihood.

With the interaction term included, the marginal effects of *VHOMO* and *SHOMO* on DVs need to account for the effects of each other. Therefore, we calculated their marginal effects as the partial derivative of the DVs as follows:

Table 3
Summary statistics of key variables.

Variable	Obs	Mean	Std. dev.	Min	Max
SUBGAIN	1622	7714.65	14,406.31	0.00	164,828.20
COOP	1622	0.12	0.12	0.00	0.98
HYPE	1622	0.06	0.08	0.00	0.80
PEER	1622	0.05	0.06	0.00	0.47
SPREAD	1622	0.01	0.03	0.00	0.33
VHOMO	1622	0.09	0.11	0.00	0.98
SHOMO	1622	0.01	0.03	0.00	0.35
DREC	1622	0.03	0.04	0.00	0.33
INDR	1622	0.09	0.10	0.00	0.98
TIER	1622	4.10	0.64	1.00	5.00
TENURE	1622	2822.93	1105.87	514.00	5611.00
STRMLEN	1622	49,033.04	38,083.11	0.00	369,232.00
avgSPECPOP	1622	263,662.80	260,907.80	18.11	695,320.40
avgR_HOSTVIEW	1622	1600.82	5150.07	11.95	83,763.75
avgR_RAIDVIEW	1622	1591.73	4674.89	14.00	54,671.50
avgR_HSUBGAIN	1622	115.89	285.81	0.00	5045.33
avgR_RSUBGAIN	1622	123.91	229.09	0.00	2836.50
GENDER	1615	0.19	0.39	0.00	1.00

Note: we standardized the independent variables before performing regression analyses.

$$H1: \frac{\partial Y_i}{\partial VHOMO_i} = \alpha_1 + \alpha_2 SHOMO_i > 0,$$

$$H2: \frac{\partial Y_i}{\partial SHOMO_i} = \alpha_3 + \alpha_4 VHOMO_i > 0,$$

when evaluated at the different values of the interaction term. We conducted multicollinearity diagnostics. The VIF values for all the independent variables are below 5, suggesting that multicollinearity is not a concern in our cooperative strategy models (see Appendix A Table A1).

4.4.2. Network effects (co-created values) of cooperation model

To test our NEs of cooperation hypotheses (H5-H7), we specified a Generalized Linear Model (GLM) using log-negative binomial regression (Log-NBR) [63] for two reasons. First, our DV, *SUBGAIN*, is a count variable with <5.56 % of the records with zero value, thus following a negative binomial distribution. Second, the log link function allows us to capture the non-linear and multiplicative effects of the different strategic cooperative events (*HYPE*, *PEER*, and *SPREAD*) on *SUBGAIN*. We specified the NEs of cooperation model as follows:

$$\log\{E(SUBGAIN_i)\} = \beta_0 + \beta_1 HYPE_i + \beta_2 PEER_i + \beta_3 SPREAD_i + \beta_4 Controls_i,$$

where $\log\{E(SUBGAIN_i)\}$ represents the logarithm of the expected value of *SUBGAIN* for initiator *i*. The coefficients (βs) represent the log-relative risks associated with changes in the independent variables - *HYPE*, *PEER*, and *SPREAD*, while controlling for other factors. Our multicollinearity diagnostic shows that the VIF values for all the independent variables are below 5, indicating that multicollinearity is not a concern in our NEs of cooperation models (see Appendix A Table A2).

5. Results

5.1. Model-free evidence

We conducted the Kolmogorov-Smirnov (KS) tests to provide model-free evidence for our NEs of cooperation hypotheses. To do so, we categorized focal streamers who, on average, initiated hyping, peering, and spreading at least once a week during the study period as frequent "hyper," "peerer," and "spreader," respectively. In Fig. 5, the Y axis shows the cumulative probability of initiators, and the X axis shows the number of new subscribers gained by streamers during the cooperative events. The KS test results suggest that focal streamers who frequently initiated hyping, peering, and spreading events acquired significantly more subscribers, compared to those who cooperated less frequently (see Fig. 5).

Co-created values. Utilizing Rustlog,[i] a Twitch logging API service, we collected the historical chat logs from the audience of our sample recipients involved in the cooperative events. This chat-log dataset contains over 94 million chat messages from over 2.67 million unique active audience (or chatters) across 663 streamers' channels during our study period. By extracting and analyzing chat messages related to the three value-related propositions (value-in-exchange, value-in-experience, and value-in-use) from these channels, we offer an innovative approach to validate the value-related propositions. Specifically, we used the ANOVA approach to compare the daily average occurrences of these value-related chats on dates with cooperative events against those without cooperative events (see Section 5.1). We demonstrate the specific processes as follows.

In order to extract chats pertinent to the three value-related concepts – value-in-exchange, value-in-experience, and value-in-use – we implemented a multi-step procedure. Initially, we conducted standard preprocessing on the raw chat logs, as outlined by Berger et al. (2020). This included cleaning data, removing stop words, tokenization, and lemmatization for text preprocessing.

For the extraction process, we utilized Word2Vec, a well-established Python-based word embedding program [64], employing its skip-gram algorithm. This approach is designed to transform words into vectors, wherein the spatial proximity of these vectors represents semantic closeness. In simpler terms, words with similar meanings or contexts are positioned closer in the vector space, facilitating the identification of contextually related terms [65].

We began with target words closely linked to our value-related propositions: "exchange," "experience," and "useful." To optimize the Word2Vec model, we employed a grid search method to identify the training windows that yield the highest similarity scores. The optimal settings determined were as follows: a window of 15 for exchange-related chats (maximum similarity score of 0.827), 10 for experience-related chats (maximum similarity score of 0.758), and 7 for use-related chats (maximum similarity score of 0.831).

During the training phase, we defined the context window as spanning up to ten words on either side of a target word. This setup allows our skip-gram model to capture the contextual nuances around our target words effectively. The model subsequently generated the top 10 keywords that are contextually similar to each target word, as demonstrated in Table 4. We then extracted chat messages containing these keywords from the cleaned chat log dataset.

Subsequently, we calculated the daily average frequency of chats pertaining to the three value-related propositions (value-in-exchange, value-in-experience, and value-in-use) for each recipient in our dataset. This calculation was performed for both days marked by cooperative events (such as hosting and raiding) and non-event days. We then employed ANOVA with repeated measures for each of the three value constructs to examine the differences in daily average chat occurrences between event and non-event days. The findings in Table 5 reveal a significant increase in chat frequencies related to value-in-experience, value-in-exchange, and value-in-use on days when cooperative events occurred, compared to non-event days. These results provide support that cooperative events among streamers facilitate value-centric dialogues among audience, which may subsequently influence streamers' market performance.

5.2. Main results

5.2.1. Results for cooperative strategy models

We adopted FLRM models to examine the associations between the

[i] https://github.com/gempir/rustlog/blob/master/README.md

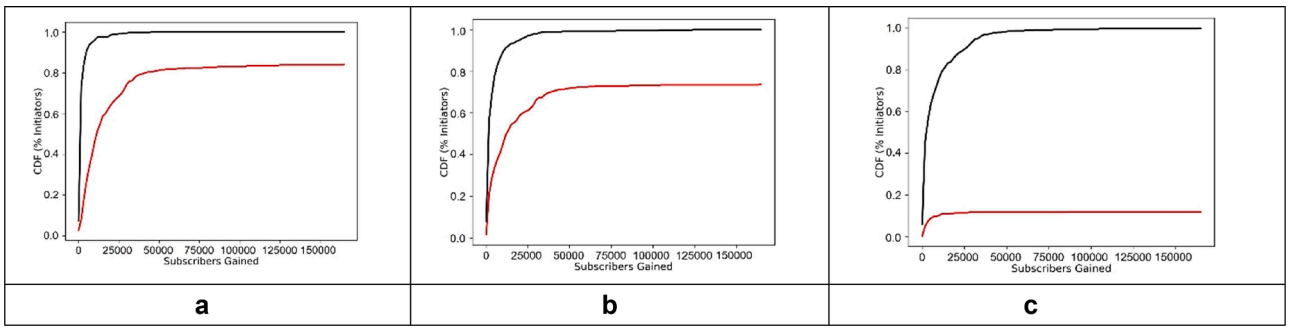


Fig. 5. a. KS Test for Hyping. Notes: The red line represents hyping with higher frequency (mean = 14,519.264, standard deviation= 18,596.787), and the black line represents hyping with lower frequency (mean= 1992.585, standard deviation= 4356.293). The KS statistic is 0.673 (*p*-value= 0.000). b. KS Test for Peering. Notes: The red line represents peering with higher frequency (mean= 12,067.120, standard deviation= 17,183.023), and the black line represents peering with lower frequency (mean= 4505.722, standard deviation= 10,865.207). The KS statistic is 0.461 (*p*-value= 0.000). c. KS Test for Spreading. Notes: The red line represents spreading with higher frequency (mean= 5534.508, standard deviation= 11,640.004), and the black line represents spreading with lower frequency (mean= 7990.102, standard deviation= 14,385.196). The KS statistic is 0.880 (*p*-value= 0.000).

Table 4
Dictionary of words related to co-created values based on Word2Vec model.

Co-created value	Top 10 most relevant keywords	# of Chats extracted	Example
Value-in-experience	enjoying, positive, hope, positive, appreciate, supporting, loving, comfy, hate, continuing	484,241	"I enjoyed the comfy stay! lilyBlanket"
Value-in-exchange	payers, subs, resub, gift, pleasegift, re-sub, resubscribed, premium, gifting, gifted	472,535	"oaklandoof gifted a Tier 1 sub to ShooterRed97!"
Value-in-use	easier, able, learned, weapon, skill, winning, armor, item, mechanic, clearly	286,705	"This is much easier to build at the spot, surely"

homophily-based and reciprocity-based cooperative strategies and streamers' cooperative behaviors. The test results are shown in Table 6 and Fig. 6. In Model 1, we tested the average effects of different cooperative strategies (*VHOMO*, *SHOMO*, *DREC*, and *INDR*) on streamers' strategic cooperation (*COOP*). We also tested the effects of the different cooperative strategies on hyping, peering, and spreading events in Model 2, Model 3, and Model 4, respectively. We calculated the marginal effects (MEs) of the different cooperative strategies and plotted in Fig. 6.

The average MEs of *VHOMO* and *SHOMO* on DVs, are measured as the partial derivatives of DVs, at different values of the interaction term *VHOMO* × *SHOMO*. As shown in Fig. 6a and Fig. 6b, the average MEs of *VHOMO* and *SHOMO* on *COOP* are significantly positive across the different values of the interaction term. Hence, our Hypotheses 1a and 2a are supported. The predictive MEs of *DREC* and *INDR* on *COOP* are also significantly positive across their different values (see Fig. 6c and Fig. 6d). Thus, our Hypotheses 3a and 4a are supported. Moreover, the results show (1) *VHOMO* has the most salient positive effect on hyping (Coeff.= 0.265, *p*-value= 0.000) (H1b is supported), (2) *SHOMO* plays less salient positive role in peering events (Coeff.= 0.051, *p*-value= 0.002) (H2b is supported), (3) *DREC* has the strongest positive effect on peering (Coeff.= 0.104, *p*-value= 0.000) (H3b is supported), and (4) *INDR* has the least positive effect on hyping (Coeff.= 0.048, p-value= 0.004) (H4b is supported).

Table 5
ANOVA analyses.

Co-created value	Date group	Mean	F-value	*P*-value
Value-in-experience	with cooperative events without cooperative events	10.730 5.428	518.017	<0.0001
Value-in-exchange	with cooperative events without cooperative events	11.100 5.477	537.894	<0.0001
Value-in-use	with cooperative events without cooperative events	8.317 2.977	728.239	<0.0001

5.2.2. Results for NEs (co-created values) of cooperation model

We show the results of Log-NBR for testing our NEs of Cooperation hypotheses (H5, H6, H7). in Table 7 and Fig. 7. Specifically, we find that the occurrence probability of hyping events (*HYPE*) exhibits a significant and positive multiplicative impact on the number of subscribers gained (*SUBGAIN*) of initiators during cooperative events, suggesting positive NEs of hyping events on initiators' market performance (H5 is supported) (see Fig. 7a). The results also show a significantly positive multiplicative impact of the occurrence probability of spreading events (*SPREAD*) on *SUBGAIN*, suggesting positive NEs of spreading events on initiators' market performance (H7 is supported) (see Fig. 7b). However, the occurrence probability of peering events (*PEER*) does not show a multiplicative impact on *SUBGAIN* of initiators (H6 is not supported). In Section 6.1, we discuss the two plausible opposing effects of peering events that may cancel out each other.

5.2.3. Robustness checks

Endogeneity Checks. In this section, we follow the approach of Venkatesan et al. [66] to examine the potential endogeneity issue in our NEs of Cooperation Model. We suspect the different cooperative events are endogenous, as the initiators' decisions to cooperate with less-, similar-, or more-popular partners may be influenced by other omitted contextual variables. Cultural norms, for instance, are shown to influence individuals' status-driven behaviors in diverse cultural environments [67]. Hence, in addition to the main dataset, we also collected self-disclosed geographical locations of our focal streamer samples from StreamsChart.com and their social media profiles (e.g., X or YouTube), as geographic locations are the closest available data that can reflect streamers' cultural backgrounds. We then created a dummy variable *GEO*, with 1 indicating North America and 0 otherwise, as our candidate instrumental variable of *HYPE*, *PEER*, and *SPREAD*.

We then adopted the Hausman test for the instrumental variable analyses as it is particularly suitable for determining the endogeneity of independent variables in cross-sectional analyses [68]. This approach

Table 6
Cooperative strategy FLRM model results.

	Model1: COOP			Model2: HYPE			Model3: PEER			Model4: SPREAD		
	Coeff.		Std. err.	Coeff.		Std. err.	Coeff.		Std. err.	Coeff.		Std. err.
VHOMO	0.310	***	0.022	0.265	***	0.017	0.181	***	0.024	0.186	***	0.041
SHOMO	0.072	***	0.013	0.068	***	0.008	0.051	**	0.018	0.066	**	0.024
DREC	0.092	***	0.017	0.048	**	0.016	0.104	***	0.023	0.089	**	0.031
INDR	0.092	***	0.015	0.095	***	0.022	0.066	***	0.017	0.079	***	0.016
VHOMO*SHOMO	-0.025	***	0.004	-0.026	***	0.002	-0.017	***	0.004	-0.025	***	0.006
TIER	0.027	**	0.010	-0.014		0.021	0.221	***	0.022	0.620	***	0.050
TENURE	0.015		0.011	0.019		0.016	-0.027		0.017	0.050		0.031
avgSPECPOP	-0.001		0.011	-0.011		0.015	-0.002		0.015	0.066	*	0.027
avgR_HOSTVIEW	-0.026		0.035	-0.070		0.040	0.017		0.026	0.092		0.065
avgR_RAIDVIEW	0.011		0.028	0.008		0.029	-0.002		0.023	0.058		0.045
GENDER	0.046		0.024	0.016		0.030	0.048		0.038	0.106		0.064
Intercept	-1.268	***	0.012	-1.509	***	0.016	-1.782	***	0.017	-2.663	***	0.050
Num of Obs.	1615			1607			1615			620		
Wald chi2	1575.700			4387.340			732.760			464.010		
Prob > chi2	0.000			0.000			0.000			0.000		
Pseudo R2	0.125			0.113			0.109			0.192		

Notes: *** $p < .001$, ** $p < .01$, * $p < .05$. Std.err. is robust standard error. Coeff. represents standardized coefficient= log odds ratio * (standard deviation of predictor/ standard deviation of the log odds ratio of dependent variable). The results for the variables of interest are bolded in the table.

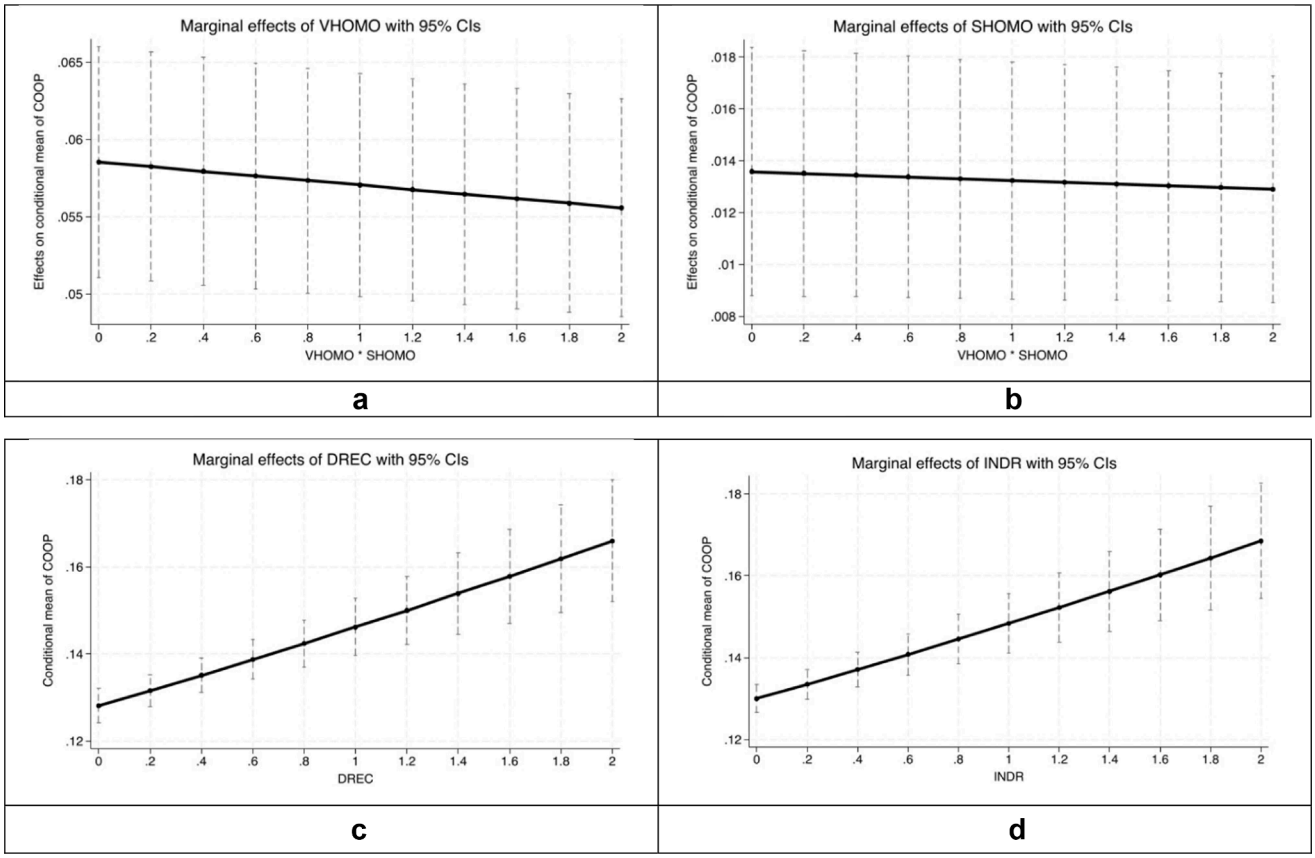


Fig. 6. a. MEs of VHOMO on COOP. b. MEs of SHOMO on COOP. c. MEs of DREC on COOP. d. MEs of INDR on COOP.

involves two-stage estimations. The first stage adopts FLRM and regresses the suspected endogenous variables (*HYPE*, *PEER*, and *SPREAD*) on our candidate instrumental variables *GEO* and estimates the residuals. The second stage uses Log-NBR and regresses the residuals on the dependent variable *SUBGAIN* and applies the F-test to test the significance of the residuals (*resid_HYPE*, *resid_PEER*, *resid_SPREAD*), where the null hypothesis suggests that exogeneity of the independent variables. Accordingly, rejecting the null hypothesis would imply the existence of endogeneity of the independent variables. The first stage of our Hausman test suggests that *GEO* are statistically significant in determining *HYPE*, *PEER*, and *SPREAD*. However, the second stage of our Hausman test fails to reject the null hypotheses, indicating that *HYPE*, *PEER*, and *SPREAD* are exogenous. For detailed test results, see Appendix B Table B1, Table B2

Sensitivity Analyses. Our dataset includes not only hosting but also raiding as forms of cooperation among streamers on Twitch. Notably, "hate raid" – a targeted attack conducted by malicious users or bots using the raiding feature – has been a significant concern since the feature's inception on the platform [69]. In response to hate-raid attacks, Twitch and its community introduced several safety measures

Table 7
NEs of cooperation log-NBR model results.

DV: SUBGAIN	Coeff.		Robust Std. err.	z	95 % conf. interval	
HYPE	0.921	***	0.133	6.940	0.661	1.181
PEER	0.082		0.082	1.010	-0.078	0.242
SPREAD	0.186	**	0.043	2.970	0.063	0.309
TIER	-0.230	**	0.088	-2.610	-0.402	-0.057
TENURE	0.014		0.059	0.230	-0.103	0.130
STRMLEN	0.946	***	0.084	11.220	0.781	1.111
avgSPECPOP	0.187	***	0.058	3.240	0.074	0.300
GENDER	0.517	***	0.144	3.600	0.236	0.799
avgR_RSUBGAIN	-0.125	*	0.051	-2.460	-0.225	-0.025
avgR_HSUBGAIN	0.084		0.116	0.720	-0.144	0.312
Intercept	8.460	***	0.060	140.530	8.342	8.578
Num of Obs.	1615					
AIC	19.114					
BIC	-2235.720					
(1/df) Deviance	2.228					
(1/df) Pearson	1.460					

Notes: *** $p < .001$, ** $p < .01$, * $p < .05$. Coeff. represents the log-relative risks associated with changes in the predictors.

from late 2021, including Shield Mode[j] and the "panic button.[k]" These tools allow streamers to better moderate their live chats and safeguard their channel environments from toxic influence. Although these safety measures have notably decreased the frequency of hate raids since their implementation, it is important to acknowledge that they could potentially influence our findings. To address this issue, we conducted sensitivity analyses that differentiated the occurrence probabilities of hosting and raiding events and examined their precursors and effects on streamers' market performance. The results of these analyses indicate that the statistical significance and direction of the four resource-based cooperative strategies on the probabilities of raiding and hosting are consistent. The impact of the three types of cooperation on the market performance of the initiators in both raiding and hosting also remains stable (refer to Appendix C). This stability underscores the robustness of our results, suggesting that even when hate raids are factored into the analysis, they do not substantially alter the overarching conclusions.

Alternative Tests. To ensure the robustness of our main results of cooperative strategy models, we further performed alternative regression analyses with zero-inflated negative binomial (ZINB) models with count variables of hyping, peering, and spreading events. The ZINB model results are largely consistent with our main results and do not disagree on the direction of effects in our main results (see Appendix D Tables D1, D2). We also measured direct reciprocity with an extended 60-day reciprocal window. The results are consistent with our main analyses (see Appendix D Table D3). Moreover, to ensure the robustness of our main results of NEs of the cooperation model, we repeated the Log-NBR with an alternative measure – follower gained during cooperative events (*FOLGAIN*). The alternative test results are consistent with our main results (see Appendix D Table D4). Furthermore, we conducted event-level analyses as alternative tests of our NEs of cooperation hypotheses, as event-level analyses allow us to control for 1) paired-level factors, such as the tier distance between initiators and their partners (*TIER_DIFF*), 2) event-level factors, such as hosting or raiding events (*EVENT_TYPE*), 3) platform-level factors, such as the number of live channels (*LIVECHA*), concurrent viewers (*PLATVIEWER*), and the number of Twitch official events (*TWITCHEVENT*), and 4) weekend or weekday influence (*WEEKEND*) (for summary statistics, see Appendix D Table D5). The results of event-level analyses are also consistent with our main results of NEs of cooperation models (see Appendix D Table D6).

[j] https://safety.twitch.tv/s/article/Protect-your-channel-with-Shield-Mode
[k] https://www.theverge.com/22633874/how-to-stop-a-hate-raid-twitch-safety-tools

5.2.4. Additional analyses

Through our main analyses, we found empirical evidence that streamers' tendency to initiate hyping and spreading events significantly influences their market performance on synchronous UGC platforms. Nevertheless, whether these positive influences may contribute to platform growth is unknown, which is critical for synchronous UGC platforms to determine the strategic approach to promoting streamers' cooperation. Therefore, we leveraged the event-level dataset and conducted a multilevel (event level and daily platform level) mixed-effects GLMs with Log-NBR analyses to examine the impacts of the different types of strategic cooperative events among streamers on platform growth (in terms of live content contributions). We specify our models as follows:

$$\log\{E(LIVECHA_j)\} = \beta_0 + \beta_1 X_j + \beta_2 Controls_j + \mu_j + \epsilon,$$

where $\log\{E(LIVECHA_j)\}$ represents the logarithm of the expected value of *LIVECHA* for cooperative event *j*. The coefficients (β_1) represent the log-relative risks associated with changes in the independent variables - *HYPE*, *PEER*, and *SPREAD*, which are dummy variables, while controlling for other variables in the model. μ_j are the random effects, and ϵ is the error term. In this model, we controlled for various factors that may influence the number of live channels on the Twitch platform. These include (1) initiator-level factors, such as initiator tenure (*TENUER*), gender (*GENDER*), streaming minutes (*STRMLEN*), and popularity tier (*TIER*), (2) initiators' specialized content popularity (*SPECPOP*), (3) paired-level factors, such as the tier distance between initiators and their partners (*TIER_DIFF*), (4) event-level factors, such as hosting or raiding events (*EVENT_TYPE*), (5) platform-level factors, such as the number of concurrent viewers (*PLATVIEWER*) and the number of Twitch official events (*TWITCHEVENT*), and (6) weekend or weekday influence (*WEEKEND*) (see summary statistics in Appendix D Table D5). As the independent variables are mutually exclusive dummy variables, we estimated their effects on DV individually to avoid collinearity. Our additional analyses revealed that hyping events have a detrimental effect on live content contributions, while spreading events have a positive effect (see Table 8). We also plotted the marginal effects of *HYPE* and *SPREAD* on *LIVECHA* in Fig. 8a and b. We discuss the theoretical and practical implications of our findings in the next section.

6. Discussion and implications

Online cooperation has become increasingly popular for bridging content creators in the competitive synchronous UGC market, where content creation has transitioned into professional online activities. Yet, little has been done to understand the underlying mechanisms that drive cooperation for this purpose and how the cooperative activities may fuel or impede creators' and platform growth. Motivated as such, we propose a holistic research framework for unveiling the resource-based cooperative strategies and the NEs that emerge from live content creators' cooperation. Our robust cross-sectional analyses provide empirical support for the majority of our hypotheses. We discuss the theoretical and practical contributions of our findings in this section.

6.1. Theoretical contributions

This study extends the literature on user engagement in UGC markets by addressing the cooperative behaviors on synchronous UGC platforms. While existing literature on synchronous UGC platforms predominantly focuses on the motivations behind live content creators' contributions and audience engagement, there remains a significant gap concerning the prevailing cooperation among live content creators. This study bridges this gap by exploring these cooperative dynamics and their economic impacts on live content creators' market performance, thereby contributing to both the resource-based view of strategic cooperation and two-sided market theories in the context of

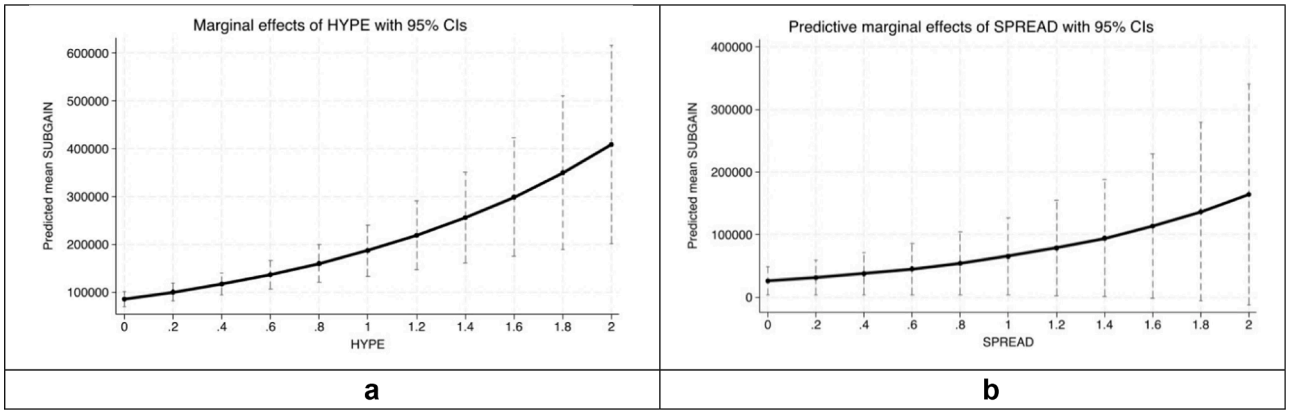


Fig. 7. a. MEs of HYPE on SUBGAIN. b. MEs of SPREAD on SUBGAIN.

synchronous UGC.

First, we integrate the resource-based view of strategic cooperation with synchronous UGC market dynamics by identifying four resource-based cooperative strategies – namely, vocational homophily, structural homophily, direct reciprocity, and indirect reciprocity – that help live content creators obtain distinct resources and drive their strategic cooperation. Our findings also detail the effects of the four strategies on the different types of strategic cooperation – hyping, peering, and spreading – categorized by the relative popularity status of live content creators. Specifically, the findings suggest that vocational homophily consistently exerts the most salient impact in driving the three types of strategic cooperation, which aligns with prior findings that creators form social ties primarily based on similarity in their content specialties [12,13]. These findings indicate that live content creators engaging in various types of cooperation prioritize their attainment and maintenance of audience and reputational resources through vocationally homophilous cooperation.

In addition, our results highlight the significance of vocational homophily and indirect reciprocity as crucial resource-based strategies for live content creators who initiate upstream cooperation. Initiators in these scenarios are likely to focus on the attainment and expansion of resources, prioritizing access to unique, non-redundant social resources [59]. In contrast, direct reciprocity is identified as less impactful in driving hyping events because it often leads to the acquisition of redundant resources, which does not facilitate resource expansion. Similarly, structural homophily is deemed the least effective strategy for creators engaged in peer cooperation (or peering). This is due to the increased direct competition faced by creators of similar popularity within the same team, which can limit the benefits derived from such cooperative strategies [33].

Furthermore, this study conceptualizes synchronous UGC markets as two-sided markets and enriches the literature by theorizing the NEs that arise from the three types of strategic cooperation—hyping, peering, and spreading—with the lens of value cocreation theory. Our study shows that both hyping and spreading events exert positive NEs on the initiators' subscriber and follower gains, indicating expansion effects of hyping and spreading at the individual level. We suggest that the expansion effect of hyping at the individual level emerges from social learning [48], access to new audience as well as diverse and higher-quality content [49,50]. We argue that the expansion effect of spreading can be a result of the reciprocal returns from the less-popular recipients [59], the reinforcement of audience commitments [50], and the enhancement of initiators' network reputation [18].

However, our additional multilevel analyses suggest that live content creators' tendency to initiate hyping events can be detrimental to live content contributions on synchronous UGC platforms. This may be due to the enlarged popularity disparity among live content creators through hyping [59], which diminishes the contribution values the platform carries to the less popular and the newcomers. The additional analyses at the platform level also provide evidence that the greater economic benefits brought by spreading events to their less-popular partners [59] may enhance new and less-popular creators' perceived emotional

Table 8
Multilevel mixed effects GLM with log-NBR model results.

DV: LIVECHA	Coeff.		Std. err.	Coeff.		Std. err.	Coeff.		Std. err.	$P > z$
HYPE	-0.006	***	0.001							
PEER				0.002		0.001				
SPREAD							0.004	***	0.001	0.001
TIER_DIFF	0.000		0.001	-0.001		0.001	-0.002	***	0.001	0.000
TIER	-0.002	***	0.001	-0.001		0.000	-0.002	***	0.001	0.001
TENURE	0.000		0.000	0.000		0.000	-0.001		0.000	0.109
STRMLEN	0.001	**	0.000	0.001	**	0.000	0.001	**	0.000	0.002
TWITCHEVENT	-0.001	***	0.000	-0.001	***	0.000	-0.001	***	0.000	0.001
SPECPOP	-0.002	***	0.000	-0.002	***	0.000	-0.002	***	0.000	0.000
PLATVIEWER	0.020	***	0.001	0.020	***	0.001	0.020	***	0.001	0.000
WEEKEND	0.108	***	0.001	0.108	***	0.001	0.108	***	0.001	0.000
GENDER	0.001		0.001	0.001		0.001	0.001		0.001	0.250
EVENT_TYPE	0.002	*	0.001	0.002	**	0.001	0.002	*	0.001	0.012
Intercept	11.432	***	0.001	11.428	***	0.001	11.429	***	0.001	0.000
/lnalpha	-5.556		0.006	-5.556		0.006	-5.556		0.006	
Num of Obs.	43,835			43,835			43,835			
Walk Chi2 (11)	29,196.820			29,127.130			29,146.220			
Prob > chi2	0.000			0.000			0.000			

Notes: Std.err. is robust standard error. Coeff. represents the log-relative risk associated with changes in the predictors. The results for the variables of interest are bolded in the table.

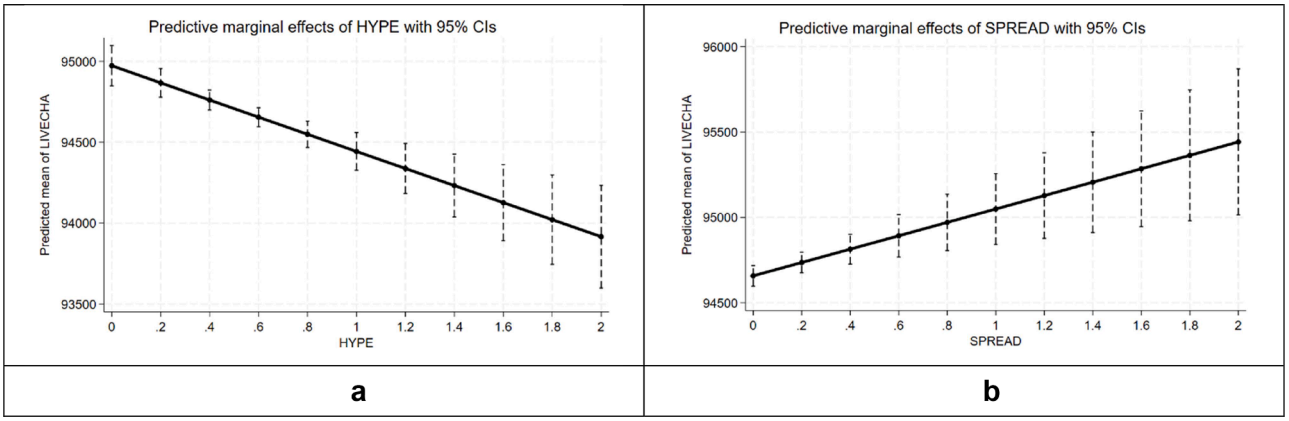


Fig. 8. a. MEs of HYPE on LIVECHA. b. MEs of SPREAD on LIVECHA.

support and their self-efficacy in competing for market shares [15], further contributing to the positive NEs of spreading on live content contribution at the platform level.

Our result of the insignificant impact of the peering events on individual market performance and live content contribution at the platform level indicates that although the peering strategy shares the added values the spreading strategy carries to the initiators, it may simultaneously suffer adverse homophilous effects in the hierarchical dimension. In other words, frequently cooperating with similarly popular creators may limit the breadth of audience network reach and the diversity in knowledge gains of the initiators [28], consequently neutralizing the impact of peering events on initiators' market performance and live content contributions.

6.2. *Practical implications*

With the increasing prevalence of content monetization, two-sided synchronous UGC platforms face significant challenges, including rising competition and growing disparity among live content creators. These dynamics risk exacerbating market imbalance, potentially leading to platform failure [9]. To address these concerns, synchronous UGC platforms have implemented various cooperative features to promote cooperation among live content creators. Nevertheless, cooperation in a competitive market can be intricate and may sometimes yield unintended market consequences.

Our findings on the conflicting effects of hyping events at individual and platform levels shed light on a potential incentive misalignment between live content creators and the platforms: while initiating hyping events greatly boosts individuals' market performance, widespread adoption of such upstream cooperation can reduce overall content contributions on the platform. To mitigate this, platform practitioners could implement mechanisms to strategically restrain the frequency or scope of hyping events. For example, the platform could introduce quota systems, where each live content creator is allotted a maximum number of hyping events they can initiate within a specific timeframe.

In contrast, our results reveal consistent expansion effects of spreading events at both individual and platform levels, positioning them as a "win-win" strategic cooperation mechanism for synchronous UGC platforms. To capitalize on this, platform managers can design features that encourage spreading events. For instance, reinforcing team identities could be achieved by introducing team-based reward systems or visible creator collectives, where collaborative groups are publicly recognized and incentivized. Additionally, enabling direct reciprocal returns could involve offering tangible benefits – such as shared audience pools, bonus revenue shares, or platform perks – for creators who participate in successful spreading events. Lastly, recommending fresh partnership opportunities for relatively more popular creators can be implemented through AI-driven matchmaking systems that identify complementary creators for potential collaboration, thereby fostering dynamic downstream cooperative networks.

7. Conclusion and future research

In this study, we examine live content creators' strategies to cooperate with others and the economic impacts of cooperation at both individual and platform levels in the synchronous UGC context. We categorize cooperation into three types – hyping, peering, and spreading – based on the relative popularity statuses of the initiators and the recipients in a cooperative event. We also theorize the NEs that arise from the cooperation among live content creators. Using data from Twitch, we uncover the different magnitudes of the resource-based cooperative strategies underlying the three types of cooperation. Moreover, our analyses also reveal the NEs originating from the three types of cooperative events, identifying the expansion effects of hyping and spreading on individual growth and the cannibalization effect of hyping on live content contributions at the platform level. Our study provides rich theoretical implications for synchronous UGC researchers as well as critical managerial implications for synchronous UGC platform stakeholders.

However, our study is not without limitations and hence opens up ample opportunities for future research on cooperative economics in the synchronous UGC markets. First, our study investigates the live content creators' cooperation by aggregating the effects of hosting and raiding events on Twitch. Future studies can include other more synchronous cooperative features, such as co-stream, and investigate the differential impacts of the different cooperation mechanisms on individual and platform growth. Second, the current study primarily focuses on the initiators' market performance to uncover the strategic consideration and market outcomes behind the interesting yet counterintuitive cooperative behaviors in competitive synchronous UGC markets. Future studies may take into account both initiators and their partners' market performance and investigate the strategic alliance performance from the different status-based cooperative events.

Third, our study concentrates on benign cooperation among streamers. Nevertheless, it is crucial not to overlook disruptive and malicious interactions, such as hate raids, which can manifest through negative reciprocity or retaliatory actions, particularly on synchronous UGC platforms characterized by intense competition. Therefore, future research should consider expanding our current framework to explore these negative interactions.

Moreover, our study focuses on the cooperative behaviors among content creators in a synchronous UGC context in which audience is allowed to engage with one channel at a time. Future studies may investigate the motivations and network effects of cooperation in asynchronous UGC settings. Furthermore, we acknowledge that while our empirical findings derive primarily from quantitative analyses, they

could be further enriched by incorporating qualitative insights. Such insights would provide a deeper understanding of individual motivations and the complex dynamics of cooperation. Therefore, we recommend that future studies employ mixed methods approaches to complement and extend the insights provided by our current research.

Last, while our dataset is robust and our analytical methods are comprehensive, the cross-sectional design limits our ability to establish causality definitively. Future research could conduct longitudinal study to provide stronger evidence for causal relationships between strategic cooperation and market outcomes on synchronous UGC platforms as well as to uncover potential strategic shifts among live content creators over time.

Submission declaration

We confirm that the manuscript titled "Online Creators' Strategic Cooperation in Two-sided Synchronous UGC Market: Empirical Evidence from A Livestreaming Platform" is an original submission that has not been published before or is currently under review at any other publication outlet. We have removed all identifying contents that could compromise a blind review.

CRediT authorship contribution statement

Bingyi Wu: Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Charles Zhechao Liu:** Writing – review & editing, Data curation, Conceptualization. **Hongyi Zhu:** Writing – review & editing, Visualization, Methodology.

Declaration of competing interest

None.

Appendix A. Multicollinearity checks

Table A1
Multicollinearity check for cooperative strategy models.

Variable	VIF
VHOMO	2.370
SHOMO	2.170
DREC	1.570
INDR	1.730
VHOMO*SHOMO	2.130
TIER	1.130
TENURE	1.140
avgSPECPOP	1.220
avgR_HOSTVIEW	2.880
avgR_RAIDVIEW	2.840
GENDER	1.190
Mean VIF	**1.800**

Table A2
Multicollinearity check for NEs of cooperation models.

Variable	VIF
HYPE	1.51
PEER	1.79
SPREAD	1.47
TIER	1.88
TENURE	1.17
STRMLEN	1.26
avgSPECPOP	1.14
avgR_RSUBGAIN	2.18
avgR_HSUBGAIN	2.00
GENDER	1.16
Mean VIF	**1.56**

Appendix B. Hausman tests for endogeneity checks

Table B1
First-stage hausman tests with FLRM.

	DV: HYPE			DV: PEER			DV: SPREAD		
	Coefficient		Std. err.	Coefficient		std. err.	Coefficient		std. err.
TIER	-0.471	***	0.069	0.420	***	0.063	1.481	***	0.161
TENURE	0.038		0.059	-0.105	*	0.048	0.092		0.083
STRMLEN	-0.421	***	0.061	-0.608	***	0.062	-0.473	***	0.131
avgSPECPOP	0.015		0.056	-0.050		0.042	-0.010		0.078
avgR_RSUBGAIN	-0.092		0.069	0.130		0.068	0.456	***	0.077
avgR_HSUBGAIN	-0.004		0.059	0.000		0.058	0.025		0.066
GENDER	0.052		0.131	0.135		0.092	0.083		0.201
GEO	-0.341	**	0.119	-0.077	**	0.033	-0.038	**	0.019
Intercept	-2.668	***	0.107	-3.076		0.085	-5.097	***	0.227
Num of obs	1616			1622			622		
Wald chi2	102.040			241.860			185.040		
Prob > chi2	0.000			0.000			0.000		
Pseudo R2	0.052			0.054			0.140		

Notes: *** $p < .001$, ** $p < .01$, * $p < .05$. Std.err. is robust standard error.

Table B2
Second-stage Hausman test with log-NBR.

DV: SUBGAIN	Coefficient	std. err.	z	$P > z$	[95 % conf.interval]	
HYPE	9.954	3.572	-2.79	0.005	-16.955	-2.952
PEER	25.125	7.960	-3.16	0.002	-40.726	-9.524
SPREAD	22.848	8.275	2.76	0.006	6.629	39.067
TIER	-1.021	0.204	-5	0.000	-1.421	-0.621
TENURE	-0.000	6.46E-05	-1.85	0.064	0.000	7.09E-06
STRMLEN	-2.45E-06	5.36E-06	-0.46	0.647	0.000	8.05E-06
avgSPECPOP	1.74E-07	2.10E-07	0.83	0.407	0.000	5.86E-07
avgR_RSUBGAIN	0.002	0.001	4.74	0.000	0.001	0.003
avgR_HSUBGAIN	-0.001	0.000	-3.96	0.000	-0.001	-0.000
GENDER	0.336	0.128	2.58	0.010	0.080	0.583
resid_HYPE	19.512	2.991	6.52	0.501	-13.650	25.373
resid_PEER	30.058	8.516	3.53	0.311	-13.366	46.749
resid_SPREAD	-15.100	9.229	-1.64	0.102	-33.188	2.988
Intercept	14.428	1.225	11.78	0.000	12.028	16.828

Notes: Std.err. is robust standard error. Coeff. represents the log-relative risks associated with changes in the predictors.

Appendix C. Sensitivity analyses

Table C1
Summary statistics of cooperative types for host (H) and raid (R).

Variable	Obs.	Mean	Std. Dev.	Min	Max
H_SUBGAIN	648	4433.79	7713.80	0.00	73,830.00
R_SUBGAIN	648	3280.86	7225.72	0.00	90,998.00
H_HYPE	1622	0.04	0.06	0.00	0.53
H_PEER	1622	0.02	0.04	0.00	0.31
H_SPREAD	1622	0.01	0.02	0.00	0.16
R_HYPE	1622	0.01	0.03	0.00	0.42
R_PEER	1622	0.03	0.04	0.00	0.32
R_SPREAD	1622	0.01	0.02	0.00	0.16
H_VHOMO	1622	0.05	0.06	0.00	0.53
H_SHOMO	1622	0.00	0.01	0.00	0.18
H_DREC	1622	0.01	0.02	0.00	0.15
H_INDR	1622	0.06	0.06	0.00	0.53
R_VHOMO	1622	0.04	0.05	0.00	0.45
R_SHOMO	1622	0.00	0.01	0.00	0.17
R_DREC	1622	0.01	0.02	0.00	0.18
R_INDR	1622	0.04	0.05	0.00	0.45

Notes: H_SUBGAIN and R_SUBGAIN are the total numbers of subscribers gained during an initiator's hosting and raiding events, respectively. H_HYPE, H_PEER, and H_SPREAD are the occurrence probabilities of hyping, peering, and spreading, respectively, when an initiator hosted their partners. R_HYPE, R_PEER, and R_SPREAD are the occurrence probabilities of hyping, peering, and spreading, respectively, when an initiator raided their partners. H_VHOMO, H_SHOMO, H_DREC, and H_INDR are the normalized vocational homophily, structural homophily, direct reciprocity, and indirect reciprocity when an initiator hosted their partners. R_VHOMO, R_SHOMO, R_DREC, and R_INDR are the normalized vocational homophily, structural homophily, direct reciprocity, and indirect reciprocity when an initiator raided their partners.

Table C2
Cooperative strategy behind hosting FLRM model results.

DV	H_HYPE			H_PEER			H_SPREAD		
	Coeff.		Std. err.	Coeff.		Std. err.	Coeff.		Std. err.
H_VHOMO	0.222	***	0.015	0.106	*	0.044	0.165	***	0.038
H_SHOMO	0.050	***	0.007	0.048	*	0.030	0.066	**	0.022
H_DREC	0.058	***	0.016	0.065	*	0.034	0.088	**	0.030
H_INDR	0.083	***	0.019	0.063	**	0.021	0.069	***	0.014
H_VHOMO*SHOMO	-0.018	***	0.002	-0.035	*	0.014	-0.025	***	0.006
TIER	0.085	***	0.021	0.417	***	0.061	0.549	***	0.066
TENURE	0.025		0.015	-0.085	**	0.027	0.041		0.030
avgSPECPOP	-0.015		0.015	-0.031		0.026	0.045		0.029
avgR_HOSTVIEW	-0.116		0.063	0.149	*	0.064	0.134		0.088
avgR_RAIDVIEW	0.031		0.039	-0.059		0.064	0.001		0.084
GENDER	0.052		0.029	0.004		0.059	0.062		0.064
_cons	-1.601	***	0.016	-2.372	***	0.053	-2.877	***	0.070
Number of Obs.	1607			1615			620		
Wald chi2	2445.970			265.010			326.400		
Prob > chi2	0.000			0.000			0.000		
Pseudo R2	0.087			0.168			0.171		

Notes: *** $p < .001$, ** $p < .01$, * $p < .05$. Std.err. is robust standard error. Coeff. represents standardized coefficient= log odds ratio * Standard deviation of predictor/ standard deviation of the log odds ratio of dependent variable.

Table C3
Cooperative strategy behind raiding FLRM model results.

DV	R_HYPE			R_PEER			R_SPREAD		
	Coeff.		Std. err.	Coeff.		Std. err.	Coeff.		Std. err.
R_VHOMO	0.185	***	0.016	0.182	***	0.017	0.170	***	0.034
R_SHOMO	0.083	***	0.012	0.035	**	0.011	0.052	*	0.021
R_DREC	0.020	*	0.007	0.104	***	0.018	0.073	**	0.025
R_INDR	0.091	***	0.022	0.040	***	0.011	0.064	***	0.013
R_VHOMO*SHOMO	-0.034	***	0.004	-0.009	**	0.003	-0.019	***	0.005
TIER	-0.141	***	0.026	0.061	**	0.018	0.589	***	0.034
TENURE	-0.002		0.025	0.018		0.017	0.055		0.028
avgSPECPOP	0.023		0.023	0.024		0.015	0.077	**	0.023
avgR_HOSTVIEW	0.041		0.027	-0.083		0.064	0.035		0.046
avgR_RAIDVIEW	-0.052		0.027	0.029		0.043	0.096	**	0.034
GENDER	-0.085		0.066	0.072	*	0.035	0.128	*	0.055
_cons	-2.263	***	0.024	-1.954	***	0.018	-2.905	***	0.038
Number of Obs.	1607			1615			620		
Wald chi2	720.960			2264.100			691.900		
Prob > chi2	0.000			0.000			0.000		
Pseudo R2	0.121			0.074			0.165		

Notes: *** $p < .001$, ** $p < .01$, * $p < .05$. Std.err. is robust standard error. Coeff. represents standardized coefficient= log odds ratio * Standard deviation of predictor/ standard deviation of the log odds ratio of dependent variable.

Table C4
NEs of cooperation log-NBR model results (by host and raid).

DV	H_SUBGAIN			R_SUBGAIN		
	Coeff.		Std. err.	Coeff.		Std. err.
H_HYPE	0.901	***	0.122			
H_PEER	0.086		0.078			
H_SPREAD	0.243	***	0.068			
R_HYPE				0.538	**	0.155
R_PEER				0.783		0.694
R_SPREAD				0.178	*	0.079
TIER	-0.323	***	0.082	-0.391	**	0.135
TENURE	-0.023		0.059	0.066		0.065
STRMLEN	0.943	***	0.086	1.031	***	0.086
avgSPECPOP	0.201	**	0.059	0.228	***	0.063
GENDER	0.558	***	0.156	0.610	***	0.160
avgR_RSUBGAIN	-0.119		0.063	-0.197	***	0.031
avgR_HSUBGAIN	0.036		0.118	-0.002		0.129
_cons	8.487	***	0.061	7.515	***	0.069
Num of Obs.	1615			1615		
AIC	17.996			17.335		
BIC	-2280.997			-2121.522		
(1/df) Deviance	2.145			2.437		
(1/df) Pearson	1.459			1.863		

Notes: *** $p < .001$, ** $p < .01$, * $p < .05$. Std. err. Is robust standard error. Coeff. represents the log-relative risks associated with changes in the predictors.

Appendix D. Alternative tests

Table D1
Summary statistics of alternative individual-level measures.

Alt. variable	Obs	Mean	Std. dev.	Min	Max
HYPE_COUNT	1622	35.156	44.933	0.000	366.000
PEER_COUNT	1622	25.622	29.877	0.000	150.000
SPREAD_COUNT	1622	6.869	17.760	0.000	151.000
FOLGAIN	1622	30,903.320	67,121.550	-22,525.500	699,798.000
FOLGAIN_abs	1622	31,175.240	66,995.490	0.000	699,798.000
FLOSS	1622	0.045	0.207	0.000	1.000
avgR_RFOLGAIN	1622	878.398	3648.955	-798.711	43,760.000
avgR_HFOLGAIN	1622	643.943	1602.365	-692.464	25,729.670

Table D2
Zero-inflated negative binomial regression model results.

	DV: HYPE_COUNT			DV: PEER_COUNT			DV: SPREAD_COUNT		
	Coefficient		std. err.	Coefficient		std. err.	Coefficient		std. err.
VHOMO	0.333	**	0.122	0.340	***	0.091	0.228	*	0.116
SHOMO	0.158	***	0.033	0.118	***	0.031	0.130		0.101
DREC	0.253	***	0.077	0.367	***	0.067	0.296	*	0.102
INDR	0.291	*	0.134	0.112	*	0.064	0.061	**	0.016
VHOMO*SHOMO	-0.076	***	0.009	-0.063	***	0.009	-0.070	***	0.021
TIER	-0.012		0.053	0.330	***	0.069	0.867	***	0.125
TENURE	0.030		0.039	-0.027		0.038	0.118		0.068
avgSPECPOP	0.011		0.036	0.026		0.035	0.151	*	0.062
avgR_HOSTVIEW	-0.048		0.092	0.044		0.094	0.204		0.193
avgR_RAIDVIEW	-0.038		0.071	-0.039		0.081	0.054		0.132
GENDER	0.160	*	0.080	0.228	**	0.080	0.120		0.157
Intercept	3.616	***	0.040	3.062	***	0.048	1.727	***	0.133
/lnalpha	-0.768	***	0.086	-0.444	***	0.078	-0.230	*	0.113
alpha	0.464		0.040	0.641		0.050	0.794		0.089
Wald chi2 (12)	327.210			266.650			121.110		
Prob > chi2	0.000			0.000			0.000		

Notes: Std.err. is robust standard error. For summary statistics of alternative count measures, please refer to Table C1.

Table D3
Cooperative strategy FLRM model results with 60-day reciprocal window.

	Model1: COOP			Model2: HYPE			Model3: PEER			Model4: SPREAD		
	Coeff.		Std. err.	Coeff.		Std. err.	Coeff.		Std. err.	Coeff.		Std. err.
VHOMO	0.310	***	0.022	0.265	***	0.017	0.181	***	0.024	0.186	***	0.041
SHOMO	0.072	***	0.013	0.068	***	0.008	0.051	**	0.018	0.066	**	0.024
DREC_60	0.092	***	0.017	0.048	**	0.016	0.104	***	0.023	0.089	**	0.031
INDR	0.092	***	0.015	0.095	***	0.022	0.066	***	0.017	0.079	***	0.016
VHOMO*SHOMO	-0.025	***	0.004	-0.026	***	0.002	-0.017	***	0.004	-0.025	***	0.006
TIER	0.027	**	0.010	-0.014		0.021	0.221	***	0.022	0.620	***	0.050
TENURE	0.015		0.011	0.019		0.016	-0.027		0.017	0.050		0.031
avgSPECPOP	-0.001		0.011	-0.011		0.015	-0.002		0.015	0.066	*	0.027
avgR_HOSTVIEW	-0.026		0.035	-0.070		0.040	0.017		0.026	0.092		0.065
avgR_RAIDVIEW	0.011		0.028	0.008		0.029	-0.002		0.023	0.058		0.045
GENDER	0.046		0.024	0.016		0.030	0.048		0.038	0.106		0.064
Intercept	-1.268	***	0.012	-1.509	***	0.016	-1.782	***	0.017	-2.663	***	0.050
Num of Obs.	1615			1607			1615			620		
Wald chi2	1575.700			4387.340			732.760			464.010		
Prob > chi2	0.000			0.000			0.000			0.000		
Pseudo R2	0.125			0.113			0.109			0.192		

Notes: *** $p < .001$, ** $p < .01$, * $p < .05$. Std.err. is robust standard error. Coeff. represents standardized coefficient= log odds ratio.
* Standard deviation of predictor/standard deviation of log odds ratio of dependent variable.

Table D4
NEs of cooperation model with follower gains.

DV: FOLGAIN_abs	Coefficient	Std. err.	z	$P > z$	[95 % conf. interval]	
FLOSS	-1.869	0.356	-5.250	0.000	-2.567	-1.171
HYPE	1.046	0.141	7.390	0.000	0.769	1.323
PEER	-0.177	0.088	-2.000	0.055	-0.350	0.003
SPREAD	0.400	0.087	4.630	0.000	0.231	0.570
TIER	-0.649	0.116	-5.590	0.000	-0.876	-0.421
TENURE	-0.214	0.068	-3.140	0.002	-0.348	-0.080
STRMLEN	0.703	0.095	7.380	0.000	0.517	0.890
avgSPECPOP	0.119	0.065	1.840	0.066	-0.008	0.246
avgR_RFOLGAIN	0.073	0.077	0.950	0.340	-0.077	0.224
avgR_HFOLGAIN	0.400	0.254	1.580	0.115	-0.097	0.897
GENDER	0.146	0.149	0.980	0.327	-0.146	0.439
_cons	9.791	0.070	139.640	0.000	9.654	9.929
Num of Obs.	1622.000					
AIC	21.607					
BIC	-2419.241					
(1/df) Deviance	1.884					
(1/df) Pearson	1.908					

Notes: Std.err. is robust standard error. Coeff. represents the log-relative risks associated with changes in the predictors. FOLGAIN_abs is the absolute value of FOLGAIN. FLOSS is a dummy variable with 1 indicating a negative value of FOLGAIN and 0 otherwise. In the main analyses, we controlled for the impacts of partners' subscriber gains on initiators' subscriber gains. Correspondingly, we controlled for the impacts of partners' follower gains on initiators' follower gains with avgR_RFOLGAIN and avgR_HFOLGAIN in this alternative model. For summary statistics of alternative measures in the model, please refer to Table C1, Table C2, Table C3, Table C4.

Table D5
Summary statistics of event-level measures.

Variable	Obs	Mean	Std. dev.	Min	Max
INIT_SUBGAIN	43,835	121.846	227.583	0.000	14,967.000
HYPE	43,835	0.520	0.500	0.000	1.000
PEER	43,835	0.379	0.485	0.000	1.000
SPREAD	43,835	0.102	0.302	0.000	1.000
TIER_DIFF	43,835	1.116	1.074	0.000	5.000
TIER	43,835	3.322	1.224	1.000	5.000
TENURE	43,835	2909.160	1092.056	529.000	5611.000
STRMLEN	43,835	365.137	240.096	-6.000	9386.000
TWITCHEVENT	43,835	0.697	0.797	0.000	3.000
SPECPOP	43,835	279,127.100	313,274.000	0.000	2,309,980.000
RCP_FGAIN	43,835	112.090	461.524	0.000	14,967.000
LIVECHA	43,835	94,696.010	8141.075	79,560.000	114,551.000
PLATVIEWER	43,835	4,223,631.000	431,495.700	3,353,923.000	6,707,070.000
WEEKEND	43,835	0.244	0.430	0.000	1.000
GENDER	43,835	0.215	0.411	0.000	1.000
EVENT_TYPE	43,835	0.592	0.491	0.000	1.000

Table D6
NEs of cooperation log-NBR models (event-level analyses).

DV: INIT_SUBGAIN	Coeff.	Std. err.	$P > z$	Coeff.	Std. err.	$P > z$	Coeff.	Std. err.	$P > z$
HYPE	0.127	0.030	0.000						
PEER				-0.242	0.230	0.078			
SPREAD							0.172	0.046	0.000
TIER_DIFF	-0.111	0.015	0.000	-0.182	0.016	0.000	-0.111	0.014	0.000
TIER	-0.707	0.017	0.000	-0.747	0.015	0.000	-0.790	0.017	0.000
TENURE	-0.021	0.009	0.017	-0.023	0.009	0.008	-0.028	0.009	0.002
STRMLEN	0.204	0.010	0.000	0.205	0.010	0.000	0.206	0.010	0.000
TWITCHEVENT	-0.004	0.008	0.616	-0.004	0.008	0.565	-0.004	0.008	0.639
SPECPOP	0.318	0.010	0.000	0.311	0.010	0.000	0.314	0.009	0.000
RCP_FGAIN	0.157	0.014	0.000	0.145	0.013	0.000	0.137	0.013	0.000
LIVECHA	0.077	0.011	0.000	0.077	0.010	0.000	0.075	0.010	0.000
PLATVIEWER	-0.017	0.008	0.041	-0.017	0.008	0.036	-0.016	0.008	0.046
WEEKEND	-0.068	0.024	0.005	-0.068	0.024	0.005	-0.068	0.024	0.004
GENDER	-0.086	0.021	0.000	-0.091	0.021	0.000	-0.102	0.021	0.000
EVENT_TYPE	-0.625	0.024	0.000	-0.637	0.023	0.000	-0.658	0.024	0.000
Intercept	4.917	0.024	0.000	5.081	0.020	0.000	4.988	0.015	0.000
Num of Obs.	43,835			43,835			43,835		
AIC	11.170			11.167			11.170		
BIC	-388,288.100			-388,438.300			-388,303.800		
(1/df) Deviance	1.827			1.824			1.827		
(1/df) Pearson	2.962			2.870			2.872		

Notes: *** $p < .001$, ** $p < .01$, * $p < .05$. Std.err. is robust standard error. Coeff. represents the log-relative risks associated with changes in the predictors.

Appendix E. Correlation matrix of key variables

Table E1
Correlation matrix of key variables.

	HYPE	PEER	SPREAD	VHOMO	SHOMO	DREC	INDR	TIER	TENURE	STRMLEN	avgSPECPOP	avgR_HOSTVIEW	avgR_RAIDVIEW	GENDER	avgR_HSUBGAIN
HYPE	1														
PEER	0.196	1													
SPREAD	-0.187	0.257	1												
VHOMO	0.687	0.704	0.363	1											
SHOMO	0.158	0.27	0.137	0.222	1										
DREC	0.338	0.539	0.246	0.336	0.205	1									
INDR	0.447	0.47	0.339	0.254	0.166	0.102	1								
TIER	-0.362	0.36	0.409	0.02	0.099	0.107	-0.02	1							
TENURE	-0.006	-0.219	-0.111	-0.196	-0.002	-0.041	-0.097	-0.124	1						
STRMLEN	-0.22	-0.362	-0.203	-0.379	-0.088	-0.17	-0.333	-0.119	0.222	1					
avgSPECPOP	0.037	-0.041	-0.056	0.021	-0.104	0.099	-0.072	-0.15	-0.135	-0.067	1				
avgR_HOSTVIEW	0.037	-0.052	-0.011	0.021	-0.047	0.024	0.016	-0.197	0.028	0.003	0.115	1			
avgR_RAIDVIEW	0.03	-0.084	-0.015	-0.001	-0.047	-0.016	-0.01	-0.228	-0.017	-0.005	0.078	0.805	1		
GENDER	-0.044	0.158	0.137	0.091	-0.033	0.149	-0.035	0.185	-0.24	-0.106	0.247	-0.045	-0.036	1	
avgR_HSUBGAIN	0.126	-0.032	-0.027	0.084	-0.027	0.034	0.053	-0.336	-0.069	-0.077	0.201	0.406	0.466	0.077	1
avgR_RSUBGAIN	0.077	-0.032	-0.029	0.046	-0.026	0.024	0.019	-0.231	0.002	-0.033	0.181	0.212	0.265	0.045	0.703

References

[1] Y. Zhang, L. Hua, Y. Jiao, J. Zhang, R. Saini, More than watching: an empirical and experimental examination on the impacts of live streaming user-generated video consumption, Inform. Manage. 60 (3) (2023) 103771. April.

[2] J. Koetsier, 2 Million creators make 6-figure incomes on Youtube, Instagram, Twitch globally, Forbes (2020). https://www.forbes.com/sites/johnkoetsier/2020/10/05/2-million-creators-make-6-figure-incomes-on-youtube-instagram-twitch-globally/.

[3] M. Zajc, The social media dispositive and monetization of user-generated content, Inform. Soc. 31 (1) (2015) 61–67. January.

[4] Y. Liu, J. Feng, Does money talk? The impact of monetary incentives on user-generated content contributions, Inform. Syst. Res. 32 (2) (2021) 394–409. March.

[5] K. Zhao, Y. Lu, Y. Hu, Y. Hong, Direct and indirect spillovers from content providers' switching: evidence from online livestreaming, Inform. Syst. Res. (2022) 1–20. August.

[6] M. Sjöblom, J. Hamari, Why do people watch others play video games? An empirical study on the motivations of Twitch users, Comput. Human Behav. 75 (2017) 985–996. October.

[7] T.K. Das, B.-S. Teng, A Resource-Based Theory of Strategic Alliances, J. Manage. 26 (1) (2000) 31–61. February.

[8] O. Hinz, T. Otter, B. Skiera, Estimating network effects in two-sided markets, J. Manage. Inform. Syst. 37 (1) (2020) 12–38. January.

[9] L. Jin, The creator economy needs a middle class, Harv. Bus. Rev. (2020). https://hbr.org/2020/12/the-creator-economy-needs-a-middle-class.

[10] Kollock, P. The economies of online cooperation. In *Communities in Cyberspace*. 1999, pp. 200.

[11] T. Song, Q. Tang, J. Huang, Triadic closure, homophily, and reciprocation: an empirical investigation of social ties between content providers, Inform. Syst. Res. 30 (3) (2019) 912–926. August.

[12] P. Bhattacharya, T.Q. Phan, X. Bai, E.M. Airoldi, A coevolution model of network structure and user behavior: the case of content generation in online social networks, Inform. Syst. Res. 30 (1) (2019) 117–132. March.

[13] X. Zeng, L. Wei, Social Ties and User Content Generation: evidence from Flickr, Inform. Syst. Res. 24 (1) (2013) 71–87. December.

[14] M.R. Johnson, J. Woodcock, It's like the gold rush": the lives and careers of professional video game streamers on Twitch.tv, Inform. Commun. Soc. 22 (3) (2019) 336–351. February.

[15] Q. Zhao, C.-D. Chen, H.-W. Cheng, J.-L. Wang, Determinants of live streamers' continuance broadcasting intentions on Twitch: a self-determination theory perspective, Telemat. Inform. 35 (2) (May 2018) 406–420.

[16] Z. Hilvert-Bruce, J.T. Neill, M. Sjöblom, J. Hamari, Social motivations of live-streaming viewer engagement on Twitch, Comput. Human Behav. 84 (2018) 58–67. July.

[17] Gros, Daniel, et al. "World of streaming. Motivation and gratification on Twitch." *Social Computing and Social Media. Human Behavior: 9th International Conference, SCSM 2017, Held as Part of HCI International 2017, Vancouver, BC, Canada, July 9-14, 2017, Proceedings, Part I 9*. Springer International Publishing, 2017.

[18] S. Bründl, C. Matt, T. Hess, S. Engert, How synchronous participation affects the willingness to subscribe to social live streaming services: the role of co-interactive behavior on Twitch, Eur. J. Inform. Syst. (2022) 1–18. April.

[19] R. Li, Y. Lu, J. Ma, W. Wang, Examining gifting behavior on live streaming platforms: an identity-based motivation model, Inform. Manage. 58 (6) (2020) 103406.

[20] Zhao, K., Hong, Y., Ma, T., Lu, Y., and Hu, Y. Raid the chat room: the effects of group size on user engagement in online synchronized communication. In *ICIS 2021 Proceedings*. Austin, Texas, 2021.

[21] S. Ye, S. Viswanathan, I.-H. Hann, The Value of Reciprocity in Online Barter Markets: an Empirical Investigation, MIS Q. 42 (2) (2018) 521. June-A12.

[22] M.A. Nowak, K. Sigmund, Evolution of indirect reciprocity, Nature 437 (7063) (2005) 1291–1298. October.

[23] I. Dissanayake, S. Nerur, J. Wang, M. Yasar, J. Zhang, The impact of helping others in coopetitive crowdsourcing communities, J. Assoc. Inf. Syst. 22 (1) (2021) 67–101. January.

[24] B. Fotouhi, N. Momeni, B. Allen, M.A. Nowak, Conjoining uncooperative societies facilitates evolution of cooperation, Nat. Hum. Behav. 2 (7) (2018) 492–499. July.

[25] M.A. Nowak, Five rules for the evolution of cooperation, Science 314 (5805) (2006) 1560–1563. December.

[26] J.E. Stets, P.J. Burke, Identity theory and social identity theory, Soc. Psychol. Q. 63 (3) (2000) 224–237.

[27] M. McPherson, L. Smith-Lovin, J.M. Cook, Birds of a feather: homophily in social networks, Annu. Rev. Sociol. 27 (1) (2001) 415–444.

[28] G. Ertug, J. Brennecke, B. Kovács, Zou Tengjian, What does homophily do? A review of the consequences of homophily, Acad. Manage. Ann. 16 (1) (2022) 38–69. January.

[29] G. Ahuja, F. Polidoro Jr, W. Mitchell, Structural homophily or social asymmetry? The formation of alliances by poorly embedded firms, Strateg. Manage. J. 30 (9) (2009) 941–958.

[30] J. Berger, A. Humphreys, S. Ludwig, W.W. Moe, O. Netzer, D.A. Schweidel, Uniting the tribes: using text for marketing insight, J. Mark. 84 (1) (2020) 1–25. January.

[31] R. Reagans, Preferences, identity, and competition: predicting tie strength from demographic data, Manage. Sci. 51 (9) (2005) 1374–1383. September.

[32] J. Nahapiet, S. Ghoshal, Social capital, intellectual capital, and the organizational advantage, Acad. Manage. Rev. 23 (2) (1998) 242–266. April.

[33] P. Block, T Grund, Multidimensional homophily in friendship networks*, Netw. Sci. 2 (2) (2014) 189–212. August.

[34] K. Roethke, J. Klumpe, M. Adam, A. Benlian, Social influence tactics in e-commerce onboarding: the role of social proof and reciprocity in affecting user registrations, Decis. Support Syst. 131 (2020) 113268. April.

[35] L.D. Molm, D.R. Schaefer, J.L. Collett, The value of reciprocity, Soc. Psychol. Q. 70 (2) (2007) 199–217. June.

[36] J.-C. Rochet, J. Tirole, Platform competition in two-sided markets, J. Eur. Econ. Assoc. 1 (4) (2003) 990–1029. June.

[37] J.-C. Rochet, J. Tirole, Two-sided markets: a progress report, Rand. J. Econ. 37 (3) (2006) 645–667.

[38] M.L. Katz, C. Shapiro, Network externalities, competition, and compatibility, Am. Econ. Rev. 75 (3) (1985) 424–440.

[39] M. Rysman, The economics of two-sided markets, J. Econ. Perspect. 23 (3) (2009) 125–143. September.

[40] L. Qiu, Q. Tang, A.B. Whinston, Two formulas for success in social media: learning and network effects, J. Manage. Inform. Syst. 32 (4) (2015) 78–108. October.

[41] A. Helkkula, C. Kowalkowski, B. Tronvoll, Archetypes of service innovation: implications for value cocreation, J. Serv. Res. 21 (3) (2018) 284–301. August.

[42] T.R. Tyler, S.L. Blader, Identity and cooperative behavior in groups, Group Process. Intergroup Relat. 4 (3) (2001) 207–226. July.

[43] Bourdieu, P. *Distinction: A Social Critique of the Judgement of Taste*. Harvard University Press, 1987.

[44] N. Levina, M. Arriaga, Distinction and status production on user-generated content platforms: using bourdieu's theory of cultural production to understand social dynamics in online fields, Inform. Syst. Res. 25 (3) (2014) 468–488. September.
[45] H. Bernhard, E. Fehr, U. Fischbacher, Group affiliation and altruistic norm enforcement, Am. Econ. Rev. 96 (2) (2006) 217–221. April.
[46] B.J. Biddle, Recent development in role theory, Annu. Rev. Sociol. 12 (1986) 67–92.
[47] M. Stallen, L.L. Snijder, J. Gross, L.P. Hilbert, C.K.W. De Dreu, Partner choice and cooperation in social dilemmas can increase resource inequality, Nat. Commun. 14 (1) (2023) 6432. October.
[48] S. Zhang, P.V. Singh, A. Ghose, A structural analysis of the role of superstars in crowdsourcing contests, Inform. Syst. Res. 30 (1) (2019) 15–33. March.
[49] Y. Park, Y. Bang, J.-H Ahn, How does the mobile channel reshape the sales distribution in e-commerce? Inform. Syst. Res. 31 (4) (2020) 1164–1182. December.
[50] T.F. Tan, S. Netessine, L. Hitt, Is Tom Cruise threatened? An empirical study of the impact of product variety on demand concentration, Inform. Syst. Res. 28 (3) (2017) 643–660. September.
[51] A.J. Hampton, A.N. Fisher Boyd, S. Sprecher, You're like me and I like you: mediators of the similarity–liking link assessed before and after a getting-acquainted social interaction, J. Soc. Pers. Relat. 36 (7) (2019) 2221–2244. July.
[52] J.M. Podolny, A status-based model of market competition, Am. J. Sociol. 98 (4) (1993) 829–872. January.
[53] H.-T. Tsai, P. Pai, Too engaged to contribute? An exploration of antecedents and dual consequences of newcomer engagement in online social groups, J. Assoc. Inf. Syst. (2021) 23.
[54] R.K. Merton, The Matthew effect in science, Science 159 (3810) (1968) 56–63. January.
[55] Z. Wang, L. Yang, J. Hahn, Winner takes all? The blockbuster effect on crowdfunding platforms, Inform. Syst. Res. (2022) 1–26. September.
[56] L. Rokach, O. Maimon, Clustering Methods, in: O. Maimon, L. Rokach (Eds.), Data Mining and Knowledge Discovery Handbook, Springer US, Boston, MA, 2005, pp. 321–352.
[57] P. Bholowalia, A. Kumar, EBK-means: a clustering technique based on elbow method and K-means in WSN, Int. J. Comput. Appl. 105 (9) (2014) 75–8887.
[58] A. Chuan, J.B. Kessler, K.L. Milkman, Field study of charitable giving reveals that reciprocity decays over time, Proc. Natl Acad. Sci. 115 (8) (2018) 1766–1771. February.
[59] D. Melamed, B. Simpson, B. Montgomery, V. Patel, Inequality and cooperation in social networks, Sci. Rep. 12 (1) (2022) 6789. April.
[60] L.E. Papke, J.M. Wooldridge, Econometric methods for fractional response variables with an application to 401(k) plan participation rates, J. Appl. Econom. 11 (6) (1996) 619–632.
[61] L.E. Papke, J.M. Wooldridge, Panel data methods for fractional response variables with an application to test pass rates, J. Econom. 145 (1) (2008) 121–133. July.
[62] S. Gallani, R. Krishnan, Applying the fractional response model to survey research in accounting, in: Havard Business School Working Paper 16, 2017. *–016*.
[63] Hardin, J.W. and Hilbe, J.M. *Generalized Linear Models and Extensions,* 2nd Edition. Stata Press, 2007.
[64] Mikolov, T., Chen, K., Corrado, G., and Dean, J. Efficient estimation of word representations in vector space. 2013. https://arxiv.org/abs/1301.3781.
[65] H.R. Rao, N. Vemprala, P. Akello, R. Valecha, Retweets of officials' alarming vs reassuring messages during the COVID-19 pandemic: implications for crisis management, Int. J. Inf. Manage. 55 (2020) 102187. December.
[66] S. Venkatesan, R. Valecha, N. Yaraghi, O. Oh, H.R. Rao, Influence in social media: an investigation of Tweets spanning the 2011 egyptian revolution, MIS Q. 45 (4) (2021) 1679–1714. December.
[67] M.K. Brady, C.J. Robertson, J.J. Cronin, Managing behavioral intentions in diverse cultural environments: an investigation of service quality, service value, and satisfaction for American and Ecuadorian fast-food customers, J. Int. Manage. 7 (2) (2001) 129–149. June.
[68] J.A. Hausman, Specification tests in econometrics, Econometrica 46 (6) (1978) 1251–1271.
[69] C. Meisner, Networked responses to networked harassment? Creators' coordinated management of "hate raids" on twitch, Soc. Media Soc. 9 (2) (2023) 20563051231179696. April.
[70] H. Li, R. Shankar, J. Stallaert, Invested or indebted: ex-ante and ex-post reciprocity in online knowledge sharing communities, ACM Trans. Manag. Inf. Syst. 11 (1) (2020) 1. January1-1:26.

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